

ID Solicitud

JLR27-00011

Tipo Solicitud

Junior Leader Retaining

Datos personales

Información general

Nombre: Demetrio**1er apellido:** Ferro**2do apellido:****País de nacionalidad:** Italy**Tipo documento:** NIE**Número documento:** Y8324577V**Género:** Male**Fecha y lugar de nacimiento**

Fecha de nacimiento: 02/12/1990**País:** Italia**Ciudad:** Sapri, Salerno**Información de contacto**

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Prefijo teléfono alternativo: Seleccione uno alternativo:

Número teléfono alternativo:

Dirección actual

País: España

Comunidad autónoma: Catalunya

Provincia: Barcelona

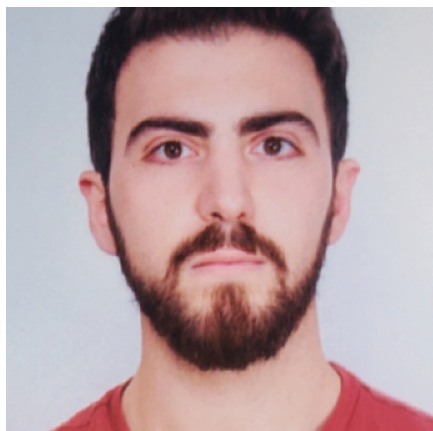
Municipio: Barcelona

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Código postal: 08003

Adjuntos

Fotografía tipo carnet



Registro de la institución anfitriona potencial

Información sobre la institución anfitriona potencial

País

042

Universidad/Centro

Universitat Pompeu Fabra - UPF

Sitio web de estudios

<https://www.upf.edu/web/cbc>

Motivación para elegir esta institución específica

The UPF is internationally renowned for its high quality research. The host lab is located at the Center for Brain and Cognition (CBC), focusing on complex functions like motor control, attention, decision-making and multisensory integration. Prof. Moreno-Bote and his group have extensively studied neural encoding and designed related theoretical models (e.g., Nogueira, et al, J. Neuroscience 2020; Kafashan, et al, Nature Commun. 2021). Colleagues at CBC (Prof. S. Soto, Prof. G. Deco) perform related human experimental work with great potential for collaboration and future developments. The CBC features state-of-the-art labs, competitive high-performance computing clusters with dedicated support, and excellent administrative facilities. Finally, Barcelona hosts a highly supportive Cognitive Computational and Systems Neuroscience (Barccsyn) community, creating an exceptional environment. Its culture of exchange via annual conferences, retreats, and seminars, makes it an ideal place.

Motivación para elegir esta institución específica

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Archivos totales: 1

Comité de selección

Área de conocimiento

Ciencias de la Vida

Disciplina

Ciencias de la Vida 3: Biología Humana, Microbiología, Biología Molecular, Genética, Biología Celular, Genómica y Proteómica, Bioquímica, Neurociencia Básica

Subámbito

Neurociencia básica

Carta de motivación

Título del proyecto

Incluya el título de su proyecto de investigación

Attention, Intention, and Free Gaze Shape Neural Selection in Laminar Sensorimotor Circuits

Resumen del proyecto

Explicación breve y concisa de su proyecto

Executive functions require the selection of relevant sensory information, intention formation, and transformation to adaptive actions.

Yet, how sensory and motor representations are selectively modulated and coordinated as behavior unfolds from sensation to action remains only marginally understood.

A crucial missing factor is gaze coordination: by simultaneously controlling where visual information is sampled and anticipating where actions are directed, it connects sensory selection, action intention and preparation.

This project will establish a multi-scale framework to determine how attention, intention and free gaze shape sensory and motor representations during an attentional hand-reach task. It will start by leveraging ~100 laminar electrophysiological sessions from M1 and PMd in two macaques performing a visually guided reaching task with free gaze and will be complemented by human EEG translation.

Aim 1 will characterize how visual features (e.g. color) and motor features (e.g. reach location) are represented and evolve from sensory selection to motor preparation in M1/PMd. Population-level analyses will test whether sensory and motor information rely on shared or distinct representations, how attentional and intentional demands enhance their encoding, and whether spontaneous gaze coordinates them. Gaze-aligned analyses will further test whether shifts toward target locations are associated with motor preparation and/or sensory reinstatement of previously displayed features.

Aim 2 will determine how these computations are organized across cortical layers, frequencies and areas. Laminar and spectral analyses will identify spectrolaminar motifs supporting sensory and motor representations in M1/PMd, testing whether principles established in visual cortex extend to motor circuits. Information-flow analyses will test hypotheses about laminar- and frequency-specific functional segregation within and between cortical areas.

Aim 3 will translate these mechanisms to humans. Healthy participants will perform the same task during EEG and eye tracking. Task variants will orthogonalize sensory selection, intended action and gaze, allowing dissociation of attentional and intentional effects. Multivariate, temporal and spectral analyses will identify human signatures of such attentional and intentional functions and their coordination with free gaze.

The central innovation is to treat free gaze as an endogenous component of executive behavior, rather than a variable to be constrained. Building on my previous work on laminar attention, gaze-dependent neural encoding and goal-directed behavior, the project will test whether gaze provides a functional bridge between visual and motor representations. The project will test whether qualitative mechanisms identified in M1 and PMd can be recovered in human EEG dynamics providing a mechanistic framework for understanding how the brain coordinates what we select, what we intend to do, and where we look.

Carta de motivación

Archivo adjunto

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Archivos totales: 1

¿Propuesta de prórroga del certificado solicitada?

No

Currículum vitae y trayectoria profesional

Identificador único del investigador/ORCID

0000-0003-4969-1415

Experiencia en investigación desde la finalización del doctorado

6

Estudios de doctorado

Nombre de los estudios

Cognitive and Brain Sciences

País

Italy

Ciudad

Rovereto

Universidad

Università degli Studi di Trento - UNITN

Nombre del director de tesis doctoral

Stefano Panzeri

Fecha de inicio

01/10/2016

Fecha fin

13/12/2019

Have you taken a career break after obtaining your PhD?

☐ Yes ☒ No

Career Break Type	Start Date	End Date	Break Days
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Otros estudios (opcional)

Nombre de los estudios	Tipo de estudios	Universidad	Expedientes académicos
Telecommunication Engineering (cum laude)	Máster	Politecnico di Torino	
Computer Engineering	Grado o equivalente	Università degli Studi di Salerno	

Empleo

Empresa u organización	Puesto de trabajo	Fecha desde	Fecha hasta
Lab-STICC, Institut Mines-Télécom	Research internship	2015-04-01	2015-09-30
Italian Institute of Technology	Research internship	2016-04-01	2016-09-30
University of Trento	PhD Student	2016-10-01	2019-09-30
Italian Institute of Technology	Research assistant	2019-10-10	2020-07-31
Universitat Pompeu Fabra	Postdoctoral researcher	2020-10-13	2025-11-30
Centre de Recerca Matemàtica	Postdoctoral researcher	2025-12-01	

Información adicional

Order number	Publication information
1.	DOI: https://doi.org/10.1109/IJCNN.2016.7727873 Title: Nearest neighbour search using binary neural networks Author(s): Demetrio Ferro*, Vincent Gripon, Xiaoran Jiang (*corresponding author) International Joint Conference on Neural Networks, IEEE, presented at Vancouver (Canada), 2016 Nearest neighbor search is a common problem in computer vision and pattern recognition, whose complexity drastically grows with data dimensionality. In this work, I developed an efficient computational method combining Product Quantization with neural networks to outperform state-of-the-art methods both in efficiency and scalability, achieving about 95% reduction in classification on MNIST dataset, and 77% reduction in nearest neighbor search on TEXMEX SIFT1M dataset.

Publication year: 2016

2. **DOI:** <https://doi.org/10.1073/pnas.2022097118>

Title: Directed information exchange between cortical layers in macaque V1 and V4 and its modulation by selective attention

Author(s): Demetrio Ferro, Jochem van Kempen, Michael Boyd, Stefano Panzeri*, and Alexander Thiele*

(*corresponding authors)

Proceedings of the National Academy of Sciences (PNAS), 2021

In this work I mapped how visual and attentional information propagate within and between cortical areas V1 and V4, introducing laminar resolution. Results show interlaminar and interareal circuit motifs, attention-related shift in the gamma-band spectral peak in V1, and increased high-frequency power in both V1 and V4. Attention reduced local communication within V1 but enhanced both feedforward and feedback signaling across areas, especially in the gamma band.

Publication year: 2021

3. **DOI:** <https://doi.org/10.1038/s41467-024-50214-2>

Title: Gaze-centered gating, reactivation, and reevaluation of economic value in orbitofrontal cortex

Author(s): Demetrio Ferro*, Tyler Cash-Padgett, Maya Zhe-Wang, Benjamin Y. Hayden, Rubén Moreno Bote

(*corresponding author) Nature Communications, 2024

Gaze behavior is strongly drawn by choice commitment during visually guided risky decision-making. Subjects looked longer at screen locations where options were currently or previously presented, and these gaze patterns were linked to the respective activation or reactivation of their corresponding value representations in the orbitofrontal cortex. The neural value-related tuning predicted upcoming choices, suggesting that gaze actively contributes to the retrieval and evaluation of internal representations during sensory reinstatement and choice commitment.

Publication year: 2024

4. **DOI:** https://2025.ccneuro.org/abstract_pdf/Ferro_2025_A_closed-loop_model_coordination_gaze_control.pdf

Title: A closed-loop model for the coordination of gaze control and decision-making

Author(s): Demetrio Ferro*, Tianming Yang, Rubén Moreno Bote
(*corresponding author) Cognitive, Computational Neuroscience (CCN), presented in Amsterdam, 2025

This work introduces a closed-loop model linking gaze behavior with value-based decision making. Visual input drives competing decision populations, which guide gaze shifts that feed back into the decision process. Simulations capture key behavioral dynamics, offering a framework to study how gaze shapes neural and decision-related activity.

Publication year: 2025

5. **DOI:** <https://doi.org/10.1038/s41467-026-70423-1>

Title: Accumulation of virtual tokens towards a jackpot reward enhances performance and value encoding in dorsal anterior cingulate cortex

Author(s): Demetrio Ferro*, Habiba Azab, Benjamin Y. Hayden and Rubén Moreno-Bote
(*corresponding author) Nature Communications, 2026

Using a token-based paradigm, we found that in proximity of a jackpot, achieved by accumulating 6 total tokens, subjects perform choices faster and more accurately, reflecting a fundamental shift in their evaluation strategy. The weight of alternative choice options in determining subjective choices is best described by a reference-dependent shift tied to the trial-based cumulative tokens count. Neural activity in dorsal anterior cingulate cortex tracked this reference-dependent shift, revealing a neural basis for reward proximity in decision-making.

Note: this publication was an invited submission by senior editor, and it has been selected in the Nature Portfolio collection "Neural mechanisms underlying sensory-guided motor control" (<https://www.nature.com/collections/deacdbghjh>)

Publication year: 2026



Publication information

DOI: <https://doi.org/10.1109/IJCNN.2016.7727873>

Title: Nearest neighbour search using binary neural networks

Author(s): Demetrio Ferro*, Vincent Gripon, Xiaoran Jiang (*corresponding author)
International Joint Conference on Neural Networks, IEEE, presented at Vancouver (Canada), 2016

Nearest neighbor search is a common problem in computer vision and pattern recognition, whose complexity drastically grows with data dimensionality. In this work, I developed an efficient computational method combining Product Quantization with neural networks to outperform state-of-the-art methods both in efficiency and scalability, achieving about 95% reduction in classification on MNIST dataset, and 77% reduction in nearest neighbor search on TEXMEX SIFT1M dataset.

Order
number

1.

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on year: 2
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DOI: <https://doi.org/10.1073/pnas.2022097118>

Title: Directed information exchange between cortical layers in macaque V1 and V4 and its modulation by selective attention

Author(s): Demetrio Ferro, Jochem van Kempen, Michael Boyd, Stefano Panzeri*, and Alexander Thiele*
(*corresponding authors)
Proceedings of the National Academy of Sciences (PNAS), 2021

2. In this work I mapped how visual and attentional information propagate within and between cortical areas V1 and V4, introducing laminar resolution. Results show interlaminar and interareal circuit motifs, attention-related shift in the gamma-band spectral peak in V1, and increased high-frequency power in both V1 and V4. Attention reduced local communication within V1 but enhanced both feedforward and feedback signaling across areas, especially in the gamma band.

Publication year: 2021



DOI: <https://doi.org/10.1038/s41467-024-50214-2>

Title: Gaze-centered gating, reactivation, and reevaluation of economic value in orbitofrontal cortex

Author(s): Demetrio Ferro*, Tyler Cash-Padgett, Maya Zhe-Wang, Benjamin Y. Hayden, Rubén Moreno Bote (*corresponding author) Nature Communications, 2024

3.

Gaze behavior is strongly drawn by choice commitment during visually guided risky decision-making. Subjects looked longer at screen locations where options were currently or previously presented, and these gaze patterns were linked to the respective activation or reactivation of their corresponding value representations in the orbitofrontal cortex. The neural value-related tuning predicted upcoming choices, suggesting that gaze actively contributes to the retrieval and evaluation of internal representations during sensory reinstatement and choice commitment.

Publication year: 2024



DOI: https://2025.ccneuro.org/abstract_pdf/Ferro_2025_A_closed-loop_model_coordination_gaze_control.pdf

Title: A closed-loop model for the coordination of gaze control and decision-making

Author(s): Demetrio Ferro*, Tianming Yang, Rubén Moreno Bote

(*corresponding author) Cognitive, Computational Neuroscience (CCN), presented in Amsterdam, 2025

4.

This work introduces a closed-loop model linking gaze behavior with value-based decision making. Visual input drives competing decision populations, which guide gaze shifts that feed back into the decision process. Simulations capture key behavioral dynamics, offering a framework to study how gaze shapes neural and decision-related activity.

Publication year: 2025



DOI: <https://doi.org/10.1038/s41467-026-70423-1>

Title: Accumulation of virtual tokens towards a jackpot reward enhances performance and value encoding in dorsal anterior cingulate cortex

Author(s): Demetrio Ferro*, Habiba Azab, Benjamin Y. Hayden and Rubén Moreno-Bote
(*corresponding author) Nature Communications, 2026

5. Using a token-based paradigm, we found that in proximity of a jackpot, achieved by accumulating 6 total tokens, subjects perform choices faster and more accurately, reflecting a fundamental shift in their evaluation strategy. The weight of alternative choice options in determining subjective choices is best described by a reference-dependent shift tied to the trial-based cumulative tokens count. Neural activity in dorsal anterior cingulate cortex tracked this reference-dependent shift, revealing a neural basis for reward proximity in decision-making.

Note: this publication was an invited submission by senior editor, and it has been selected in the Nature Portfolio collection "Neural mechanisms underlying sensory-guided motor control"
(<https://www.nature.com/collections/deacdbghjh>)

Publication year: 2026

Research achievement

First and corresponding author of multiple high-impact journal articles (including Nature Communications 2026 and 2024 and PNAS 2021) and peer-reviewed conference proceedings (Cognitive Computational Neuroscience CCN, 2023 and 2025; IJCNN, IEEE, 2016).

Recently submitted a preprint to bioRxiv (2026), ready for submission to a high-impact journal.

Recipient of competitive funding (BIAL Foundation, 2022) supporting my postdoctoral research.

Strong interdisciplinary profile spanning systems and computational neuroscience, psychology, physics, and engineering.

Expertise in information theory, statistical modelling, data analysis, and software development.

Research Contributions

Laminar and spectral organization of cortical communication and the effects of attention

A central goal of systems neuroscience is to understand how information is selectively routed across cortical circuits. During my doctoral research, I developed information-theoretic and spectral methods to characterize directed interactions within and between cortical layers in brain areas V1 and V4, focusing on covert attentional selection. This work provided a laminar- and frequency-resolved account of inter-areal laminar communication under attention, showing that attentional selection modulates information flow through rhythmic coordination, affecting local laminar processing and broad inter-areal exchange. These findings established a mechanistic link between cortical microcircuit structure, oscillatory dynamics, selective information routing, and attention.

Representative publications:

Ferro, Boyd, van Kempen, Panzeri, PNAS, 2021

Ferro, Institutional Research Information System, 2019

Gaze and internal states in value-based decision-making

My first postdoctoral work, supported by BIAL 2022, extended the principles of attentional selection from sensory processing to value-based decision-making. I established active gaze coordination as a core component of preference formation, rather than simply a readout of choice. Building directly on my earlier framework of attentional routing, I developed a broader view of attention as a multi-level computational principle linking information acquisition, neural representation, and decision dynamics. In this framework, gaze is an overt expression of decision-related attentional shifts that dynamically interacts with internal representations. My work showed that gaze modulates orbitofrontal value representations and supports their online reactivation and reevaluation during decisions, establishing a link between attention, gaze control, neural computation, and economic decision-making.

Representative publications:

Ferro, Cash-Padgett, Zhe-Wang, Hayden, Moreno-Bote, Nature Communications, 2024

Ferro, Rifé-Mata, Cash-Padgett, Zhe-Wang, Hayden, Moreno-Bote, CCN, 2023

Reference-dependent reward value computation in decision-making circuits

I subsequently extended this framework to internal goal drives, asking how reward structure changes the computations supporting choice. My recent work showed that proximity to a large reward systematically reshapes valuation dynamics, improves behavioral performance, and enhances value-related neural encoding in the dorsal anterior cingulate cortex. These findings demonstrate that internal motivational context is not an overall modulatory factor, but a structured component of decision computation that



shapes how prospective value information is represented and used. Together with my work on gaze, this research shows how both active sampling and internal state can dynamically reconfigure neural representations during goal-directed behavior.

Representative publications:

Ferro, Azab, Hayden, Moreno-Bote, Nature Communications, 2026

Ferro, Azab, Hayden, Moreno-Bote, CCN, 2025

Computational models of neural computation and decision-making

Throughout my career, I have developed computational models to understand how information is represented, processed, and transformed to behavior. My work has combined information theory and neural networks, from neuro-inspired algorithms for high-dimensional nearest-neighbor search to recurrent and closed-loop models linking sensory evidence, gaze allocation, and decision formation. I have recently developed computational frameworks to investigate dynamic interactions between sensory and decision-making circuits, showing how their recurrent interactions can shape the temporal weighting of sensory evidence. Together, these approaches provide mechanistic frameworks linking information processing, neural dynamics, active sensing, and behavior.

Representative publications:

Ferro, Gripon, Jiang, IJCNN, 2016

Ferro, Yang, Moreno-Bote, CCN, 2025

Ferro, Vivar, Levi, Huk, Wimmer, bioRxiv, 2026

These works have been presented at 30+ international venues.

My website provides detailed, updated lists of presentations (<https://d-ferro.github.io/research>) and publications (<https://d-ferro.github.io/publications>), also available through Google Scholar.

Peer recognition

Invited and contributed presentations at leading international research institutions and scientific networks

These invitations reflect recognition of my expertise in attention, gaze, decision-making, neural information processing, and computational neuroscience.

All of these invitations came from peer researchers conducting related research programmes and with whom I shared methods, software code, mutual insights and conceptual discussions.

List of main invited talks:

2026 MSCA-COFUND TOUCH, Parc de Recerca, Univ. Autònoma de Barcelona (Barcelona, Spain)

2025 Computational and Mathematical Neuroscience, by Dr. A. Hyafil (Barcelona, Spain)
2024 Swammerdam Institute for Life Sciences, by Dr. C. Bosman (Amsterdam, Netherlands)
2023 Center for Mind/Brain Sciences by Prof. R. Bottini (Rovereto, Italy)
2022 Multi-Scale Brain Function India-Italy Network of Excellence by Prof. G. Naldi (Varenna, Italy)
2020 Institut de Neurosciences de la Timone by Dr. B. Kilavik and Dr. A. Brovelli (Marseille, France)
A full list of 25+ contributed presentations (talks/posters) at international conferences is available on my website (<https://d-ferro.github.io/research>)

Competitive research funding and participation in internationally funded programmes

Independently conceived, wrote, and secured BIAL Foundation Research Grant PT/FB/BL-2022-106 (<https://www.fundacaobial.com/en-GB/grants/approved-projects/grants-2022>) for a research programme on gaze-centered decision-making, supporting two years of my postdoctoral research at Universitat Pompeu Fabra (UPF).

My postdoctoral research was additionally supported by the Howard Hughes Medical Institute (HHMI International Research Grant 55008742), the EU Human Brain Project (FET Flagship, Grant 945539) awarded to Prof. R. Moreno-Bote, and the Spanish Ministry of Science (PID2020-114196GB-I00, awarded to Prof. R. Moreno-Bote; PID2023-147627OB-I00, to Dr. K. Wimmer).

My PhD research was supported by the U.S. National Institutes of Health (NIH R01 NS108410, U19 NS107464U19) and the Simons Foundation (SFARI Explorer Grant 602849), awarded to Prof. S. Panzeri, Wellcome Trust (Grant 093104) and Medical Research Council (Grant MR/P013031/1), awarded to Prof. A. Thiele.

Together, these experiences demonstrate sustained engagement in internationally competitive research programmes, alongside a demonstrated capacity to secure independent competitive funding.

Peer review for leading journals and conferences

Invited to review manuscripts for Nature Neuroscience, Nature Communications, PLOS Computational Biology, and Qeios (verifiable at ORCID), peer reviewer for conference abstracts at Cosyne and proceedings at Cognitive Computational Neuroscience (CCN). These invitations recognize my expertise in systems and computational neuroscience, neural data analysis, laminar electrophysiology, and decision-making.

Editorial Recognition

My work about "Accumulation of virtual tokens towards a jackpot reward enhances performance and value encoding in dorsal anterior cingulate cortex" was invited for submission by Senior Editor at Nature Communications. After the publication, in March 2026, the article was also selected as part of the Nature Portfolio collection "Neural mechanisms underlying sensory-guided motor control"

(<https://www.nature.com/collections/deacdbghjh>), remarking its recognition and impact within the field of investigations.

Academic Teaching, Supervision and Examination

I taught Scientific Communication at UPF in 2021 and 2022, supervised students at bachelor and PhD stages of career, conceived and mentored their thesis projects.

I chaired Biomedical Engineering B.Sc. thesis examination committees at UPF in 2024–2026 and an M.Sc. thesis committee in 2022.

I was invited to serve as Chair of two PhD thesis defense committees at SISSA (Trieste, Italy) and as referee for a PhD thesis at UPF in 2026.

These roles reflect recognition of my scientific expertise and maturity as an independent researcher.

International Scientific Collaborations

I established, led and sustained research collaborations with internationally recognized groups at Baylor College of Medicine and UCLA (USA), NYU-ECNU (China), Institute for Advanced Simulation Jülich (Germany), and Aix-Marseille University (France), Newcastle University (UK). These collaborations have resulted in multiple first corresponding-author publications, including two Nature Communications papers (2024, 2026), PNAS (2021), CCN (2023, 2025) and span expertise in gaze, decision-making, laminar electrophysiology, neural computation, and large-scale neural data analysis.

I contributed to the Human Brain Project in the "Consortium of Research in Decision Making" combining theoretical (PIs: R. Moreno-Bote, I. Cos from Barcelona, A. Desthèze from Paris) and experimental (PIs: S. Ferraina from Rome, T. Theys from Leuven) labs. I organized and presented at scientific coordination meetings and contributed to the final collaborative report writing.

Información adicional

Interrupciones en la carrera profesional, trayectorias profesionales diversas y acontecimientos importantes de la vida

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Archivos totales: 1

Información adicional

Interrupciones en la carrera profesional, trayectorias profesionales diversas y acontecimientos importantes de la vida

I did not take a formal career break; however, my career progression was significantly impacted during 2020. After completing my PhD in December 2019, I was only able to start my postdoctoral position in October 2020. Although I had a verbal agreement with Prof. Rubén Moreno-Bote to join his lab at Universitat Pompeu Fabra, I was unable to secure a work contract without being physically present in Spain to obtain a Número de Identificación de Extranjero (NIE). This requirement was initially unfeasible due to pandemic-related lockdown, and was subsequently long delayed by the inability to schedule an appointment through the online system.

Otras contribuciones a la comunidad investigadora

Social involvement

During my PhD, I organized a yearly series of weekly seminars, and a Lab Fair to showcase science in local labs.

During my postdoc, I extensively contributed to the Barcelona Computational, Cognitive and Systems Neuroscience community (<https://barccsyn.org>) by organizing two retreats (2022 and in 2024) involving 60+ people, consisting in 3 days of fully immersive scientific and community-related activities, fostering mutual exchange and networking.

I promoted inclusive participation across genders and career stages.

I actively participate to the Barccsyn network in weekly journal clubs, monthly seminars, annual conferences.

Attracted by the "Mes Cajal" events, I established connection with the Societat Catalana de Biologia and the Societat Catalana de Matematica.

I joined the "PostDoc Day in Biomedical research", and extended invitation to neuroscience colleagues. I contribute to Centre de Recerca Matemàtica activities, including periodic meetings series of "Neurochats", where I also presented in May 2025.

Open Science

I am fondly committed to open science practices by releasing data and code and publishing research with open access licenses.

I introduced an open science and copyright licensing module in the Scientific Communication course I taught at UPF in 2021-2022.

Citizen Science

I joined the citizen science CASES network, within the EUTOPIA Impact and Dissemination programme at UPF. Initiatives cover dissemination, environmental data sampling, collective reports, and inclusion of underrepresented social groups in scientific research. I contributed by presenting my research on executive functions, lecturing about "the wonders of the brain" to school students, and microplastics sampling in Malgrat de Mar (Spain).

Media Outreach

I have published informative articles in several renowned outlets (ErurekAlert!, Bioengineering, Medical Xpress, Medical News), UPF (2021, 2024, 2026) and CRM News (2026).

Full CV in additional documents.

Documentación adicional

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- [Descargar](#)

Archivos totales: 1

Mobility check

Date from	Date to	Country	Institution, if applicable	Description
2015-04-01	2015-09-30	France	Institut Mines-Télécom	I developed my Master Thesis during a 6-months excellence internship program supervised by Prof. Claude Berrou, and Prof. Vincent Gripon. The work I conducted was published and presented at the International Joint Conference on Neural Networks (IEEE) in Vancouver, Canada in 2016.

2016-04-01	2016-09-30	Italy	Italian Institute of Technology	I started a research internship in the lab of Prof. Stefano Panzeri, supervisor of my PhD project.
2016-10-01	2019-09-30	Italy	Center for Mind/Brain Sciences, University of Trento	PhD candidate in the Dept. of Mind/Brain Sciences. Defense date 13/12/2019.
2017-01-11	2017-01-14	United Kingdom	Newcastle University	Visiting PhD student in the experimental visual neurophysiology lab of Prof. A. Thiele with whom I was collaborating during my PhD.
2017-06-11	2017-06-17	United Kingdom	Newcastle University	Visiting PhD student in the experimental visual neurophysiology lab of Prof. A. Thiele with whom I was collaborating during my PhD.
2017-12-11	2017-12-16	United Kingdom	Newcastle University	Visiting PhD student in the experimental visual neurophysiology lab of Prof. A. Thiele with whom I was collaborating during my PhD.
2018-05-13	2018-06-12	United Kingdom	Newcastle University	Visiting PhD student in the experimental visual neurophysiology lab of Prof. A. Thiele with whom I was collaborating during my PhD.
2019-10-01	2020-07-31	Italy	Italian Institute of Technology	Research Associate in the lab of Prof. Stefano Panzeri. Writing the PNAS 2021 manuscript, preparing my PhD project data and code for open access publication.

2020-08-01	2020-10-12	Spain	Unemployed	This period of time overlapped with the outbreak of COVID19. I did not intentionally take a break but I was unable to start a new work contract due to the inability to get a Spanish NIE. I verbally had agreed with Prof. Rubén Moreno-Bote about working in his lab since May 2021 but I could only start a work contract after I got the paperwork done in October 2021.
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2020-10-13	2025-11-30	Spain	Center for Brain and Cognition, Universitat Pompeu Fabra	Postdoctoral researcher in the lab of Prof. R. Moreno-Bote. I conceived and led research lines about gaze-mediated and goal-directed influences on decision making, through international collaboration with Prof Hayden (Baylor College of Medicine, USA), productive outcome (CCN 2023-2025, Nature Communications 2024-2026), conferences and invited talks. I contributed to Human Brain Project (co-funded by EU) in the Consortium of Research in Decision-Making with theoretical (PIs: R. Moreno-Bote, I. Cos, Spain; A. Desthèze, France) and experimental labs (S. Ferraina, Italy and T. Theys, Belgium). Further collaborations include T. Yang (NYU-ECNU, China) on gaze-decision coordination CCN 2025, and S. Grün (Jülich, Germany) and T. Brochier (Marseille, France) on gaze-to-motor-action coordination.
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2025-12-01	2026-09-23	Spain	Centre de Recerca Matemàtica	Postdoctoral researcher in the lab of Dr. Klaus Wimmer. Here I started to work on a new research line about computational modelling of the underlying principles of temporal evidence weighting in perceptual decision-making. For this project I established new, exciting collaboration with Prof. A. Huk (UCLA, Los Angeles, USA), combining my skills in advanced analysis of behavioral and electrophysiological data with my engineering background to design a computational architecture that emulates observed sensory and decision-making neural signatures observed in the data, while accurately predicting behavioral choice reports. Despite joining the lab relatively recently, a full draft of the manuscript is available in preprint version, ready for submission to high-impact journal.
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**Universitat
Pompeu Fabra**
Barcelona

Rubén Moreno Bote

Serra Húnter Full Professor and ICREA Academia Member
Director of the Theoretical and Cognitive Neuroscience lab
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To whom it may concern,

With the present document, I express my support to Demetrio Ferro's candidacy for the Junior Leader postdoctoral fellowships program because of three main motivations. First, Demetrio is an exceptionally talented scientist, as demonstrated by his high impact research track. Second, his project proposal is truly at the cutting edge of current advances in its field. Third, his recent work highlights his remarkable ability to dissect complex, detailed datasets, which makes him an ideal candidate for this fellowship and positions him to make significant contributions to current understanding of neocortical organization and function.

In addition to his scientific strengths, Demetrio has demonstrated remarkable scientific independence and academic maturity. He has independently conceived and led research projects, formulated original research questions, brilliantly supervised students, and driven studies from experimental design through data analysis, manuscript preparation, and publication. He has also contributed to securing competitive funding, including support from the BIAL Foundation (2022), demonstrating his ability to help attract external resources. Beyond his own research, he is an active member of the scientific community, serving as a reviewer for leading journals in the field. His expertise is further recognized through frequent invitations to present his work at national and international scientific meetings. Over the past years, he has also co-organized multiple research events within Barcelona's neuroscience community, contributed regularly to local seminars and meetings, and played a leading role in fostering international collaborations.

In the event that he would be awarded this fellowship, our institution would be interested in offering him a position. I am aware that this declaration of interest does not bind any of the parties and that, in the case that the candidate obtains the fellowship, he may change the center chosen at the time of the start of the fellowship.

Rubén Moreno-Bote

RESEARCH EXPERIENCE – My research integrates computational modeling and experimental data analysis to investigate **neural selectivity, laminar circuitry, and gaze-aligned neural coding** underlying **visual attention, decision-making, and motor control**. Across these domains, I investigate how neural units and circuits process information to uncover the computational mechanisms underlying **executive function and behavior**.

I have **led** multiple **first and corresponding author** publications in leading **journals** in the field (*PNAS*, [2021¹](#); *Nature Communications*, [2024²](#), [2026³](#)) establishing **original contributions**, demonstrating **scientific rigor, international recognition, leadership and independence**. My work spans generalizable computational models and specialized pipelines for **extracting information from complex, heterogeneous data**, with variability arising from physiological noise, behavioral states, and anatomical specificity. I have **addressed core field challenges** including the **retinotopic alignment** of multisite recordings in the visual cortex, and the **sparseness** of spiking activity in frontal brain regions, covering **diverse cognitive demands** in non-human primates (NHPs).

I have **independently conceived, secured funding for, and led** research across **all stages**, from **writing research projects** to developing **my own theoretical frameworks, supervising research**, and leading **first-, corresponding-author** manuscripts from **conception to publication**. I combine leadership with **international collaboration**, promoting **transparency and reproducibility** through **open-access** publications and by sharing data and code⁴ in line with **FAIR** principles. My work has been presented at **30+ international conferences** and in recurrent **invited talks**, fostering **visibility** and the exchange of **methods, results, insights and perspectives**. I also actively contribute to **scientific networking, community-building, citizen science, and public outreach**.

Beyond research, I have **supervised** multiple **bachelor's theses, research interns and PhD students, taught Scientific Communication (with positive evaluations)** at Universitat Pompeu Fabra (UPF) in 2021–2022 and 2022–2023, and **served as a reviewer** for manuscripts at *Nature Neuroscience*, *Nature Communications*, *PLOS Comp. Biology*, *Cosyne*, *CCN* and *Qeios* (verifiable at [ORCID](#)), reflecting **international recognition**. I have been **external referee for 3 PhD theses**, and **invited Chair of bachelor's, master's and PhD thesis committees**.

Overall, these activities have demonstrated my **independence, mentoring, collaboration, and scientific communication skills**, reinforcing my commitment to **scientific integrity and inclusivity**. Looking ahead, I aim to establish an independent research programme investigating how **fundamental neural principles** identified in NHP studies **translate to human executive functions**, providing core mechanistic insights while relying on non-invasive recording techniques. This work has broad impact on **basic science**, and on the advancement of **human biomarkers of executive function**, with fundamental future implications for **cognitive rehabilitation, clinical neuroscience, and neurotechnology**. By integrating **theoretical models, mechanistic systems neuroscience, and human research**, this project positions me for **career advancement** toward my objective to establish a **mechanistic account of neural computation** bridging **basic and translational neuroscience**.

With a background in **information theory**, I became interested in how principles of natural computation can inform artificial systems. My master's thesis in Telecommunications Engineering (*cum laude*) was conducted during a six-month EU excellence internship in the lab of Prof. Claude Berrou (Brest, France), inventor of **turbocodes** and pioneer in the field. I designed **neuro-inspired methods**⁵ for efficient nearest neighbor search, published and presented at the *IEEE International Joint Conference on Neural Networks*⁶ (Vancouver, 2016).

– **PhD contributions** I pursued my PhD in cognitive and computational neuroscience (2016–2019) in the lab of **Prof. S. Panzeri**, renowned for his expertise in applying information theory to investigate the underlying principles of neural coding. I broadened my training with courses in **Medical Neuroscience, Machine Learning, research ethics and methods (diploma transcript)**. I **independently organized a year-long seminar series** that brought together local and visiting scientists at the **Center for Mind/Brain Sciences** (Univ. of Trento, Italy). My research focused on how **visual attention** shapes **laminar interactions** within and between **visual areas V1**

and **V4**, in collaboration with [Prof. A. Thiele](#) (University of Newcastle, UK). I applied my engineering skills to build **custom analysis tools** for NHP data acquired using multi-contact probes in acute sessions. I **extracted** local field potentials (LFPs), reflecting mesoscopic-scale dynamics (<300 Hz), and multi-unit activity (MUA), related to neural spiking (300-6000 Hz). I developed a [graphical user interface](#) (GUI) to facilitate assessment of **data quality** and to **align probe depths** across sessions using [MUA latencies](#) and [current-source-density \(CSD\)](#) derived from LFPs. I **introduced an attentional laminar framework** integrating prior theories about **laminar- and spectral-specific** pathways to provide an **integrated assessment** of the **effects of attention** on **laminar microcircuit interactions**. I developed methods to map **directed flows** across laminar depths in V1 and V4 based on conditional Granger causality and Shannon's mutual information, designed to **discard indirect inter-laminar interactions**. Results were **published** in [PNAS \(2021\)](#)¹, with related [data, GUI and code](#)⁴, presented at [Society for Neuroscience](#) (Washington, 2017; Chicago, 2019) and invited talks (cited 70 times as of Sept. 2026).

- **PostDoc contributions** During my postdoctoral research in the lab of [Prof. R. Moreno-Bote](#) at UPF (2020-2025), I consolidated my independence by **leading multiple projects, supervising students, teaching Scientific Communication**, and securing **independent funding** in 2022 through a [Bial Foundation](#) grant (2023-2024). A central line of my independent research, in collaboration with [Prof. B. Hayden](#) (Baylor College of Medicine, Houston, USA), investigated how **expected reward value** is encoded in the **orbitofrontal cortex** during probabilistic **decision-making**. I designed **neural encoding** analyses time-locked to **free gaze shifts**, revealing that NHPs strategically direct their gaze to reward-cue locations, either **during cue presentation**, or later, **during blank-screen delay** epochs, to **selectively gate** or **re-activate** the encoding of expected reward value. This work was **published** in [Nature Communications \(2024\)](#)², with [data and code](#)⁴ openly shared, presented at several conferences ([SENC, Lleida 2021](#), [Cosyne Lisbon 2022](#), [MSBFIINE Milan 2022](#), [HBP Paris 2023](#), [HBP Barcelona 2023](#), [CCN Oxford 2023](#), [ICT Milan 2024](#)) and in invited talks. This work contributed to the [Human Brain Project \(HBP\)](#), was featured on [EBRAINS](#) and in [collaborative HBP report](#), in which I was directly involved, within the Consortium of Research in Decision Making, bringing together consistent theoretical (PIs: R. Moreno-Bote, I. Cos from Barcelona, Spain; A. Desthene from Paris, France) and experimental research (PIs: S. Ferraina, Rome, Italy; T. Theys, Leuven, Belgium). Building on this work, I independently conceived and led the analysis of **dorsal anterior cingulate cortex activity** during **token-based decision-making**, showing that **goal-directed influences enhance both decision speed and accuracy** while shaping the encoding of **expected outcomes** for **potential gains and losses**. The work was published in [Nature Communications \(2026\)](#)³, following **invited submission by a Senior Editor** and subsequent selection for the Nature Portfolio collection ["Neural mechanisms underlying sensory-guided motor control"](#), highlighting its relevance to the field. Committed to **reproducible science** practices, I released the related [data and code](#)⁴. The findings were also presented at conferences, including proceedings publication in [CCN Amsterdam 2025](#) and in **invited talks**.

I further expanded my international network through collaborations with [Prof. T. Yang](#) (Shanghai, China), [Prof. S. Grün](#) (Jülich, Germany) and [Dr. T. Brochier](#) (Marseille, France), extending my programme to **active visual sampling** and **sensorimotor functions**. With Prof. Yang, I developed a **closed-loop model** of **free gaze sampling** and **decision formation** ([CCN 2025](#))⁶. Ongoing analyses of NHP data collected in Prof. Yang's lab will provide a critical **test of the model** and its **theoretical predictions**. In collaboration with Prof. Grün and Dr. Brochier, I investigated the coordination of **vision, gaze and arm movements** in NHPs, identifying the **onset of gaze-mediated motor preparation activity** and **distinct neural representations** of movement segments **dynamically organized to encode** composite motor sequences ([interim report](#); in preparation).

More recently, in December 2025, I joined [Dr. K. Wimmer](#)'s lab at the Centre de Recerca Matemàtica (CRM), further extending my theoretical framework in collaboration with [Prof. A. Huk](#) (UCLA, Los Angeles, USA). I

have now fully developed a **dynamical-systems model based on coupled sensory- and choice-tuned units**, using **NHP neural and behavioral data** to set model parameters and test predictions about how **hierarchical circuits with recurrent mutual inhibition** integrate and weight **visual motion evidence** in perceptual decision-making. The work was presented at [S-in-A, Alicante 2026](#), [FENS Barcelona 2026](#), [INCTN Trieste 2026](#), and in invited talks. A preprint version⁷ is available, the manuscript is ready for submission to a leading journal.

A **full list of invited/contributed talks** is available in my **Biographical sketch** (additional documents) and on [my personal page](#). In addition, I actively contribute to **scientific community-building** and **public engagement**. I joined the [Sociedad Española de Neurociencia \(SENC\)](#) and the [Federation of European Neuroscience Societies \(FENS\)](#), I participate in events by [SfN](#), [SENC](#), [FENS](#), [CCN](#), the [Societat Catalana de Biologia \(Mes Cajal, Postdoc Day 2024\)](#) and [Societat Catalana Matemàtica](#). I extensively contribute to the [Barcelona Computational Cognitive Systems Neuroscience \(Barccsyn\)](#) community by organizing whole network retreats in [2022](#) and [2024](#) (60+ people) and presenting in annual conferences (2021–2026). I was invited as a speaker at the [CRM](#) and regularly [contribute](#) to its [NeuroChats](#). I enthusiastically engage in **citizen science** ([PA-MAP CASES](#), part of [UPF EUTOPIA Impact and Dissemination](#), with activities including school lectures, environmental monitoring sampling, and talks open to the public), interviews at **scientific magazines** ([EurekAlert!](#) [AAAS](#), [News Medical](#), [Bioengineer.org](#), [Medical Xpress](#), [UPF News 2021](#), [2024](#), [2026](#), [CRM News](#)), **social media visibility** ([Bluesky](#), [LinkedIn](#)) and **broader outreach events** ([EU Researcher's Night](#), [Festa de la Ciència](#) and [Pint of Science](#)).

- **Future vision** Through my previous research, I established **strong, cohesive** expertise in **visual attention** and **decision-making**, with a particular focus on **spectral** and **laminar** circuitry and on how **gaze** shapes **attention** and **goal-directed processes**. Leveraging a rich laminar electrophysiology dataset from [Dr. B. Kilavik](#) (Institut de Neuroscience de la Timone, Marseille), this project will bring together my expertise into a **multi-scale, mechanistic account of executive function**, grounded in **single-neuron, population, temporal, laminar, spectral, and inter-areal dynamics**. By characterizing how **attention, intention, motor preparation, and freely moving gaze** are represented and coordinated within and across cortical circuits, I will establish a cross-species, cross-scale framework for the **integration of cognition and action** and their **functional organization**.

A further objective is to determine whether these **mechanistically resolved principles in NHPs generalize to human executive function**. This objective will be addressed in collaboration with [Prof. B. Hayden](#) (Baylor College of Medicine, BCM, Houston, USA), with whom I have established and led **productive and sustained research partnerships** throughout my postdoctoral career (Ferro et al., *Nature Communications* 2024², 2026³). This will establish a **cross-species, cross-scale framework** in which circuit-level mechanisms generate testable predictions for distributed human neural activity and behavior. This approach moves beyond descriptive comparisons between species to ask which **fundamental principles of executive computation are conserved across neural systems**, and which aspects of their organization can be captured non-invasively in humans.

The project is **well aligned with the objectives of the UPF host lab**, where Prof. Moreno-Bote's expertise in **computational neuroscience** provides a strong framework for integrating the proposed mechanistic models with experimental evidence. Combined with a collaborative network spanning **NHP and human research** and dedicated [High Performance Computational](#) facilities, the UPF environment will provide an ideal setting for an independent programme at the interface of **cognitive, computational, and translational neuroscience**.

I am fully committed to establishing an independent research programme linking **laminar microcircuits to distributed human brain networks**, and **fundamental mechanisms** of cognition in NHPs to **clinically and technologically relevant non-invasive human** applications. In the long run, I aim to build a research group that combines **rigorous mechanistic investigation, cross-species validation, and human translation**, while fostering a **highly collaborative** environment grounded in **scientific integrity, inclusivity, and transparency**.

SUMMARY OF THE PROPOSED RESEARCH PROJECT – Recent advances in recording techniques have shifted the research interest towards investigating the neural mechanisms underlying executive functions at **single-cell resolution**, across **cortical layers** and **brain regions** in NHPs. In parallel, **non-invasive techniques** such as Electroencephalography (EEG) increasingly enable complementary investigations in humans. Together, these approaches create a unique opportunity to understand how **attentional** and **intentional drives** enhance the selective encoding of **sensory** (e.g., color/shape) and **motor** (e.g., target location/angle) **features**, how these neural representations are organized across **cell populations** and **cortical circuits**, and how models based on NHP electrophysiological data translate to human **executive functions**. A crucial, yet largely unexplored aspect is the role of **freely moving gaze**, which links sensory sampling, spatial attention, and goal-directed action. Although related investigations have been conducted in motor regions, previous studies have primarily relied on **experimentally instructed gaze**, often trying to minimize its effect as a spurious source of noise. Instead, allowing gaze to **unfold spontaneously**, thereby **moving beyond** constrained gaze paradigms that impose predetermined fixation or saccade patterns, provides a **more naturalistic framework** to reveal **endogenous attention-intention interactions** and test whether underlying mechanistic principles **translate to humans**.

Building on my previous experience in laminar and spectral interactions in visual cortex¹, gaze-centered encoding of action-decision variables^{2,3}, and dynamical circuitry^{6,7}, **I aim to establish a unified framework** to investigate attentional, intentional, and gaze-related neural dynamics across NHP and human circuits. This will advance models of neural computation for **cognitive focus** and **action**, reveal principles of **gaze-dependent cortico-cortical** and **frequency-specific intra- and inter-areal interactions**, providing **mechanistic foundations** for **future translational approaches** to **executive dysfunction**, **neurotechnology** and **rehabilitation**.

This project will consist of two complementary lines of research. The first will analyze novel, **unpublished NHP data** collected in full by collaborators in the lab of Dr. B. Kilavik (Institut de Neurosciences de la Timone, Marseille). The second will investigate related mechanisms in **human EEG and eye-tracking data** to be collected by the lab of Prof. B. Hayden (Baylor College of Medicine, Houston, USA), extending the mechanistic framework from NHP neural circuits to human executive function. The NHP dataset is **already preprocessed**, substantially reducing risk and allowing to **focus directly on the proposed analyses**. The acquisition of human data is **planned at the start of the project timeline**, with analyses of EEG data planned to begin at the start of the second year. At least **two fully led publications** are expected from this research (see **Gantt chart**).

- Introduction

Executive functions enable the flexible selection of relevant information and the transformation of internal goals into adaptive actions. Their disruption is highly prevalent, with attentional disorders affecting up to 11% of children⁸ and nearly 7% of adults⁹. Yet, despite their central role in behavior, the neural mechanisms dynamically connecting **sensory information, attention, intention, and action** remain poorly understood.

Recent advances in large-scale and laminar electrophysiological recordings now provide an unprecedented opportunity to investigate these processes across **single neurons, cortical layers, frequencies, and brain regions** in NHPs. Previous research, including my own contribution¹, has demonstrated that functionally distinct information pathways can be dissociated through their **spectral and laminar organization**. However, it remains unexplored how these principles extend beyond visual sensory areas to distributed circuits linking **perception and action**, and how they are dynamically shaped by **attentional and intentional demands**.

A further crucial but underexplored factor is **the role of gaze**. By continuously linking sensory sampling with spatial selection and goal-directed action, gaze provides a natural mechanism through which **attention and intention interact** across sensorimotor circuits. Understanding these interactions requires moving beyond experimentally constrained gaze and investigating how neural representations evolve under more naturalistic

conditions where subjects can **freely direct their gaze** during the execution of cognitive functions.

Finally, establishing the broader relevance of these mechanisms requires determining whether signatures identified at the circuit level in NHPs can be **translated to human systems**. Combining mechanistic insights from laminar recordings with **EEG and eye tracking** offers a unique opportunity to identify scalable markers of the interactions between **attention, intention, gaze, and action**, ultimately advancing our understanding of executive function and **dysfunction**, with impact in **basic and cross-species, translational neuroscience**.

- State of the art

Within the context of endogenous visual attention, neural focus is driven by top-down signals from higher-level structures in a hierarchical, distributed model of the neocortex, guiding lower-level computation and optimizing outcomes¹¹. In the visual cortex, top-down attention enhances sensory processing¹², reduces inter-neuronal correlations¹³, reshapes coding properties¹⁴, increases high-frequency power and coherence of LFPs¹⁵. In my previous work¹, I integrated models of **laminar**^{16,17} and **spectral specificity**¹⁸ to map directed laminar interactions in V1 and V4, revealing that **attention strengthens** input propagation and **attenuates** interactions within V1, while **enhancing** flows within V4 and between V1 and V4 in the gamma band (>25 Hz).

Recent findings by Dr. B. Kilavik's group have detailed functional separation of **motor preparation** and **visuospatial attention** in M1 and PMd LFPs, respectively related to lower (12–20 Hz) and higher (21–29 Hz) beta spectral bands¹⁰. Previous studies showed that M1 and PMd spiking activity shows partially **overlapping sensory and motor neural encoding properties** rather than selective tuning, with PMd responses being closely linked to attentional shifts^{19–20}. Crucially, the **spectral and laminar characterization** of microcircuitry in these areas is only just emerging^{21–23}, and is particularly compelling given the evidence that dopaminergic inputs mainly influence deep-layer circuits involved in reward processing, whereas serotonergic modulation mostly affects superficial processing and attentional functions^{24,25}. In addition, M1 and PMd share connections with the posterior parietal cortex (PPC), a critical hub for sensorimotor integration and motor planning^{26–28}.

Although distinct circuits are associated with spatial integration²⁹ and oculomotor control³⁰, **freely moving gaze** provides a common factor linking sensory and motor processing³¹: through movement, eyes coordinate **parafoveal** and **foveal** focus to enhance visual acuity³² and anticipate actions, improving both speed and accuracy of reaching movements³³. My recent work² shows that free gaze **gates reward-value encoding** in the orbitofrontal cortex, extending its influence **beyond sensory sampling to goal-directed decision formation**. Similarly, PMd activity is modulated by gaze during motor preparation^{34,35}, supporting both attentional and intentional functions³⁶. Previous studies showed that motor preparation is best mapped to an eye-centered reference frame in PPC, and to a limb-centered frame in PMd³⁷. Yet, whether **sensory selection** and **action intention** constitute **dissociable** computations, and how **free gaze coordinates**^{38,39} them, remains unknown.

The **translational relevance** of these mechanisms can be further investigated using **human EEG**, capturing oscillatory and event-related dynamics. Combined with **eye tracking**^{40,41}, EEG rhythms provide markers of **attention, motor preparation, executive control, and sensory integration**. Modulations in **alpha and beta bands** have been associated with attentional selection and motor preparation^{42,43}, while **theta interactions** reflect executive control and cognitive integration⁴⁴. Analyses of **oscillatory bursts, functional connectivity, and cross-frequency interactions** can bridge NHP circuit mechanisms and **distributed human brain dynamics**, providing scalable signatures of **attention, intention, gaze, and sensorimotor** interactions.

- Objectives

The project will pursue **three integrated aims**, combining **mechanistic investigations** in NHPs with **direct translation to human psychophysiology**. These aims will leverage **unpublished behavioral and laminar electrophysiological recordings** from M1 and PMd collected by Dr. B. Kilavik's group, and **human experiments**

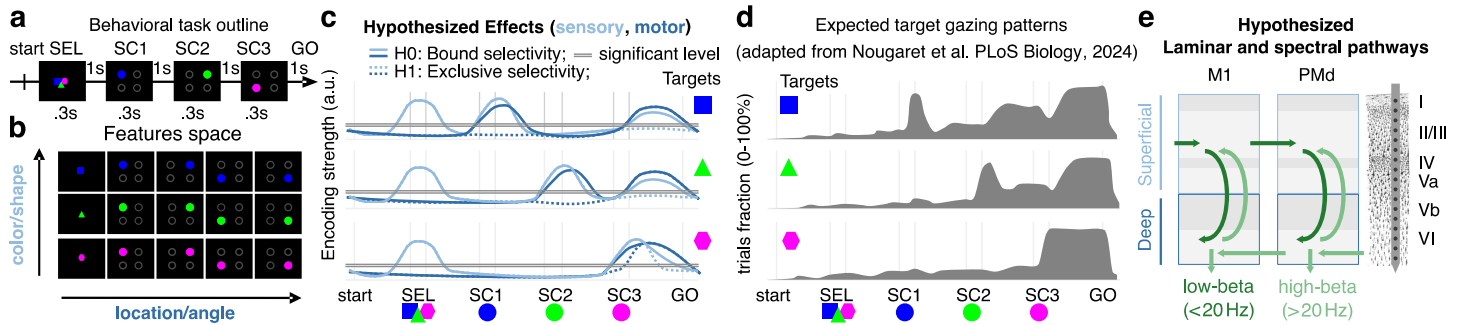


Figure 1. Task outline, hypothesized neural selectivity effects, gaze position and hypothesized laminar microcircuitry.

a. Task outline: SEL cue indicates the color to attend, 3 Spatial Cues (SC) indicating colored target locations. SEL cues are randomized. The colors of SC1–3 are fixed, locations are randomized. **b.** Sensory (color/shape) vs motor (location/angle) features. **c.** Bound vs Exclusive selectivity hypotheses, with activation at different task times. Sensory could be stronger at the time of SEL, SC and before GO; motor could be stronger at target SC and before GO. The magnitude of these could provide insights about joint or exclusive selectivity. **d.** Fractions of trials with gaze on the possible targets⁹. **e.** Hypothesized model of laminar and spectral segregation for M1 and PMd based on literature^{21–23}. Illustration of motor cortex histology (Wikimedia).

to be conducted by collaborators in the lab of [Prof. B. Hayden](#) starting at project initiation. For the NHP component, two macaques (1 male, 1 female) performed a **visually-guided hand-reaching task**, in which a selection cue (SEL) indicated the color to attend among 3 subsequent spatial cues (SC1–3), indicating the target location to reach after the GO signal (**Fig. 1a**). Correct reaches were rewarded with fluid. The setup comprised monitor, manipulandum and eye tracking. **Free gaze** is critical here, as task gaze constraints could interfere with the **spontaneous** attention–action coordination. Recordings targeted left M1 and PMd, contralateral to the reaching arm. Acute sessions used [V-Probes](#) (24–32 contacts, 0.1–0.2 mm spacing). All procedures complied with EU and French regulations (Directive 2010/63/EU, CEEA 071; Authorization 03383.02, Institution B1301404) detailed in Nougaret et al., 2024¹⁰, where these subjects performed the task for different research aims. **Fully preprocessed** NHP data include ~100 sessions including LFPs (<300 Hz), and spiking data (0.3–6 kHz): single-unit activity (SUA; ~500 in subject 1, ~1100 in subject 2), or multi-unit (MUA).

Core questions: Can sensory and motor information be dissociated as distinct neural computations, and how are they organized across M1/PMd layers, areas and timescales? Does free gaze coordinate their selection?

The human dataset will be collected by the lab of Prof. B. Hayden and will include a sex-balanced cohort of at least 30 healthy participants performing the hand-reaching task (**Fig. 1a**) with EEG and eye tracking, using a computer mouse. The human task paradigm will enable rapid implementation of task variants that would be considerably more demanding to train in NHPs. We will introduce **three targeted variants**: **(1) sensory–action decoupling**, by adding a spatial GO cue after SC1–3 that instructs reaching to a fourth, previously uncued location, dissociating the selected sensory feature from the intended reach; **(2) selection competition**, by presenting two stimuli of same color, allowing participants to freely reach either one, dissociating attentional selection from a uniquely determined motor report; and **(3) gaze control**, by comparing otherwise identical trials under free gaze and instructed fixation, testing how gaze constrains sensory and motor representations.

Core questions: Can we dissect attentional and intentional effects, and how are they coordinated with freely moving gaze across timescales—from local laminar circuitry in NHPs to distributed human brain dynamics?

– **Project timeline** (see **Risk and contingency plans**; **Impact**; **Gantt chart**)

Year 1 will focus on comprehensive NHP data analyses (Aims 1–2, detailed below), while human data are collected in parallel at the Hayden lab. NHP analyses will be accelerated by the availability of preprocessed data and by methods established through my previous work. **Year 2** will focus on human data analysis (Aim 3) and the first **Q1 manuscript** on NHP findings. **Year 3** will integrate NHP and human results into a second **Q1 manuscript**, and publication of **one proceedings paper** at [CCN 2029](#), while efforts will focus on securing research funding and progressing to a faculty position. **Dissemination** will continue throughout the 3 years in

annual [Barccsyn](#) and [SENC](#) meetings and at several international conferences: [Cosyne 2028 \(Lisbon\)](#), [FENS 2028 \(Glasgow\)](#), [SfN 2028 \(San Diego\)](#), [CCN 2029 \(Europe\)](#), and further editions based on funding availability.

- **Specific aims and methodology**

Aim 1: Characterize selective encoding and its modulation by attention, intention, and free gaze.

Main outcomes: Determine whether sensory and motor information is represented through **bound or exclusive selectivity** in the tuning of neural responses and determine their temporal evolution across sensory selection and motor preparation. Establish how neural representations are shaped by **attentional selection** and/or **action intention**, and investigate the role of **spontaneous gaze dynamics** in their coordination.

Using an **attentional match-to-sample task** (Fig. 1a) allows to dissociate the representation of independent sensory and motor features along **visual** (color/shape) and **action** (target location/angle) dimensions (Fig. 1b). I will use **regression-based analyses**^{2,3} of pre-extracted SUA to quantify feature **encoding strength** by testing linear relationships between firing rates and task features. These analyses will test our **hypotheses regarding bound versus exclusive selectivity** (Fig. 1c) and will be complemented by **non-linear information-theoretic methods**³, **representational similarity analyses**⁴⁴, and analyses of shared trial-to-trial variability, including **noise correlations**, to characterize population-level interactions of visual and action representations, leveraging a decade of expertise and scientific exchange in the labs of Prof. Panzeri⁴⁵ and Prof. Moreno-Bote⁴⁶.

Next, I will examine how **attentional** and **intentional drives** modulate neural selectivity by contrasting the encoding strength of **cued** and **non-cued** visual and motor content. The leading hypothesis is that sensory encoding is enhanced during SC1–3 **presentation**, and before the GO signal, potentially reflecting **perceptual recall**. In parallel, motor activation possibly arises during SC1–3, when **target direction is specified**, and before the GO signal, during **motor preparation**. These partially overlapping periods of input-without-output or output-without-input will provide powerful windows for principal component analyses, to identify sensory and motor representational subspaces and characterize attentional and intentional modulatory effects^{47,48}.

Lastly, I will align neural encoding to **spontaneous gaze shifts** occurring after spatial-cue presentation and before the GO signal, when SCs are no longer visible (Fig. 1d). This analysis will establish whether changes in sensory and motor representations are systematically associated with **gaze allocation**, directly linking neural selectivity to naturalistic attention–action coordination, providing key insights for executive functions.

Aim 2: Characterize and extend laminar- and spectral-specific neural organization and dynamics.

Main outcomes: Determine whether sensory and motor representations segregate across **cortical depth** and **frequency**. Identify **spectro-laminar motifs** supporting their processing. Test whether **superficial** and **deep** compartments differentially support sensory input and motor output, and how these signals interact within and between M1 and PMd (Fig. 1e). Extend established sensory **microcircuit models** to pre-/motor areas.

The probe and laminar depths will be aligned across sessions using MUA [latencies](#) and [CSDs](#), using methods established in previous work¹. I will characterize the **distribution of LFP power** and test whether the **spectrolaminar organization motif** described in **visual areas**⁴⁹ is reflected in the frequency-specific organization of premotor and motor rhythms. I will then directly test whether low- and high-beta activity (<20 and >20 Hz) show distinct laminar distributions, in addition to the column-wise segregation between M1 and PMd¹⁰.

Based on functional²² and structural²³ evidence, I will test whether superficial and deep laminar compartments show **functional specificity**, hypothesizing preferential contributions to **sensory input** and **motor output** processing, respectively (Fig. 1e). These hypotheses are motivated by functionally separable PPC inputs from areas 5a and 7d to M1 and PMd⁵⁰, together with evidence for distinct attention and motor functions in PMd and M1 LFPs¹⁰. To probe **laminar specificity**, I will contrast encoding strength for visual and motor features in

superficial versus deep SUA encoding from Aim 1 and map LFP **spectro-laminar microcircuitry** with methods previously applied to V1-V4 directed flows¹, now advanced to extract **feature-specific information transfer**⁵¹. Combining LFP and SUA analyses will enable a clearer mapping of functional network organization⁵². I will then compare cued versus non-cued feature encoding to evaluate **attentional** and **intentional modulation** across layers. These analyses will be combined with **gaze data**, integrating models of **gaze-to-action coordination**⁶, which I developed for decision-making, informing their extension to also incorporate **attentional** signals.

Aim 3: Translate NHP mechanisms of attention, intention, and free gaze to human EEG

Main outcomes: Determine whether the sensory and motor representational dynamics identified in NHPs can be paralleled in **human EEG**, and identify neural signatures that selectively reflect **sensory and motor processing, attentional selection, action intention, motor preparation, with gaze-locked coordination**.

EEG data will undergo 0.5–100 Hz band-pass and 50-Hz notch filtering, bad-channel detection/interpolation, and artifact rejection using [EEGLAB](#) and [MNE-Python](#), and time-locked to spontaneous gaze patterns. Analyses will include time-resolved regression and representational analyses to characterize **feature-coding structure**. Even though I do not expect **sex-specific aspects** in the studied functions, I will include sex as a covariate in all analyses to directly assess eventual differences. I will test whether the dissociable signatures of **attention and intention** are embedded within **oscillatory network interactions**. I hypothesize that attentional selection is reflected in spatially selective **alpha modulation**^{53,54}, whereas motor intention or preparation is reflected in **beta-band dynamics**, given the involvement of beta rhythms in sensorimotor functions^{55–57}. I will first use **sensory-action decoupling** to test whether alpha and beta modulation follows sensory selection and intended action, respectively, independently of their alignment. When the two are aligned, little additional control is required; when they diverge, the brain must maintain and reconcile competing task representations. This is where I hypothesize **theta dynamics to engage**, consistent with the established role of frontal theta in cognitive control across distributed brain regions^{58,59} and large-scale, cross-frequency dynamics⁶⁰. These predictions will also be tested in task variants with **increased selection competition** or **imposed gaze fixation**, keeping all other task components unchanged. Spectral power, gaze-aligned transient oscillatory bursts, cross-frequency coupling, and functional connectivity will therefore be used not simply to characterize EEG dynamics, but to test **which oscillatory mechanisms distinguish selection, intention, and their interaction**. The resulting human signatures will be compared with the **representational, temporal, and spectro-laminar motifs identified in M1 and PMd of NHPs**, without assuming a direct anatomical correspondence between scalp EEG and these regions. Instead, the cross-species analysis will test whether the **circuit-level principles** identified from laminar NHP recordings, at the levels of representation, temporal dynamics, and spectral organization, are recovered in EEG recordings of distributed human brain activity. This approach will establish whether mechanisms identified at the laminar microcircuit level generalize across species and recording scales, identifying candidate non-invasive signatures of **attention, intention, gaze behavior, and their interaction**.

- **Risks and Contingency plans** SUA encoding strength may vary across task features; where appropriate, I will use MUA to track amplitude encoding, an equally valid approach in this context⁶¹. Complementary strategies include spectral coherence between SUA and LFPs, leveraging functional segregation of LFPs already present in these areas¹⁰. Laminar alignment may be less straightforward because of the scarcity or absence of a granular layer⁶², typically used as reference in sensory areas. This is not a major concern, as the primary goal is to distinguish superficial from deep compartments, which show clear firing rate differences, with deeper layers exhibiting stronger activity²³. Although the proposed research program is ambitious, each aim builds on **methods and pipelines already fully developed and extensively validated in prior work** (e.g., revealing attentional/intentional effects^{1–3}, mapping laminar microcircuitry¹, quantifying encoding strength and

dissecting gaze patterns in depth^{2,3}, modeling gaze-to-action coordination⁶ and temporal evidence weighting⁷). For the human data component, **data quality** will be **closely monitored from the first acquisition sessions**, with protocols and **high quality control assessments** designed to ensure that the data meet the expected standards from the outset. As an additional safeguard, I have agreed with Prof. S. Soto to **collaborate** with his research group at UPF, **my selected host institution**, to fully implement the same **experimental protocols if required**, ensuring complementary expertise and continuity in data acquisition. Together, these strategies **ensure feasibility** offering an ideal playground to advance **novel, impactful insights** into **executive functions**.

- Impact and career development

The project will have direct impact on basic science. NHP data will generate mechanistic predictions about the physiology of executive functions, while human data will test whether such principles generalize. In the longer term, these mechanisms may inform approaches to **attentional**⁶³ and **motor**⁶⁴ **impairments**, rehabilitation, and **brain-machine interfaces**^{65,66} and **neuroprosthetics**^{67,68}. Integrating laminar circuit models with emerging **laminar neuromodulation**^{25,69} may further inform future clinical and pharmacological strategies. Moreover, the **free-gaze paradigm** provides a much **more ecologically valid** framework, opening to real-time applications like **neurofeedback**⁷⁰. Ongoing work in Dr. Kilavik's lab enables future **extension to test the origin of PPC projections**, while Hayden's lab at BCM provides access to **relevant clinical populations** (e.g., ADHD).

For **my career**, this fellowship will be **critical** for consolidating **independence**, pursuing **competitive funding** (e.g., ERC StG) and transitioning to a **group leader position** by its end. It will allow me to integrate expertise in computational neuroscience, laminar electrophysiology, gaze behavior, decision-making, statistical modeling, and dynamical systems into a **cohesive programme**, expanding into **human neuroscience** and developing **collaborations**, a **funding record**, **mentoring experience**, and **scientific visibility** to establish a **research group**.

Gantt Chart Timeline of Aims and milestones with expected execution time

Aim 1. Characterize the selectivity of neural encoding and attention, intention, free gaze.

Aim 2. Laminar and spectral specificity and extension of laminar microcircuitry models

Aim 3. Collection of data in Prof. Hayden's lab (Baylor College of Medicine, Houston, USA)

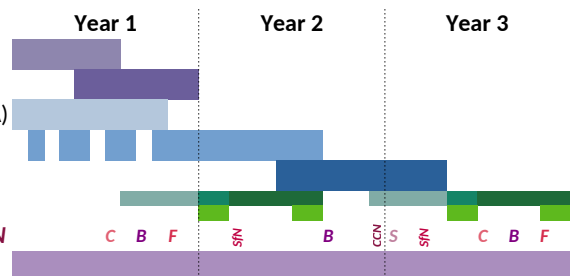
Aim 3. Active monitoring of data quality, preprocessing, visual/motor feature encoding.

Aim 3. Attentional/intentional/gaze dissections, cross-frequency coupling effects.

Publications (manuscript **preparation/submission/reviewing**, code/data **release**)

Conferences in Spain Barccsyn(**B**), SENC(**S**) and international Cosyne(**C**), FENS(**F**), SfN, CCN

Progress reports (Group/department presentations, online meetings with collaborators)



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UNIVERSITÀ
DI TRENTO

Repubblica Italiana / *Italian Republic*
in nome della Legge / *in the name of the Law*

Noi, Professor Paolo Collini, Rettore dell'Università degli Studi di Trento,
visti gli atti del compimento del corso e il risultato dell'esame finale superato in data 13 dicembre 2019
relativi al Dottorato di Ricerca in Cognitive and Brain Sciences
Curriculum Cognitive Neuroscience

*We, Professor Paolo Collini, Rector of the University of Trento,
given the successful completion of the prescribed programme of study and research and
the outcome of the final examination held on December 13, 2019
at the Doctoral Programme in Cognitive and Brain Sciences
Cognitive Neuroscience Curriculum*

conferiamo a / *confer upon*

DEMETRIO FERRO

nato a Sapri (Salerno, Italia) il 2 dicembre 1990 / *born in Sapri (Salerno, Italy) on December 2, 1990*

il titolo di / *the degree of*

DOTTORE DI RICERCA / DOCTOR OF PHILOSOPHY

in

COGNITIVE AND BRAIN SCIENCES

COGNITIVE AND BRAIN SCIENCES

Il presente diploma è valido a tutti gli effetti di legge
This certificate is a legally valid document

Il Rettore / *The Rector*

BOLLO ASSOLTO VIRTUALMENTE
AUTORIZZAZIONE N. 106668/14 D.D. 23.12.2014 RILASCIATA DALL'AGENZIA DELLE ENTRATE DI TRENTO

TIPOGRAFIA DELL'UNIVERSITÀ DEGLI STUDI DI TRENTO
TRENTO 20/02/2020

EDUCATION AND STUDENT SERVICES MANAGEMENT

We hereby certify that Mr. **DEMETRIO FERRO**, born in Sapri (Salerno, Italy) on December 2, 1990, attended regularly all the formative activities of the Doctoral Programme in “Cognitive and Brain Sciences” 32nd cycle, with administrative seat at the University of Trento (Italy).

The 32nd cycle of the PhD Programme in “Cognitive and Brain Sciences” lasts three years (from the 1st of November 2016 to the 31st of October 2019) and the PhD candidates are requested to attend all the training activities.

On December 13, 2019, Mr. DEMETRIO FERRO successfully completed the final requirement of the Program, by defending his dissertation and was granted the title of “Dottore di Ricerca in Cognitive and Brain Sciences - curriculum Cognitive Neuroscience” (Ph.D. in Cognitive and Brain Sciences - Cognitive Neuroscience curriculum).

As per Law 183/2011, art. 15, this certificate cannot be submitted either to an Italian public body or to private administrators of public utilities services in Italy.

Trento, 24th March 2020

Divisione Supporto Corsi di Dottorato e Alta
Formazione - Polo Città e Rovereto
Fabrizia Patton

UNIVERSITÀ DEGLI STUDI
DI TRENTO



DOCTORATE IN COGNITIVE AND BRAIN SCIENCES

ACADEMIC TRANSCRIPT

Name of Institution: Doctorate in Cognitive and Brain Sciences Center/University: Center for Mind/Brain Sciences / University of Trento (ITALY) Tel.: +39 0464 808617 Fax: +39 0461 288690 E-mail: phd.cimec@unitn.it Administrator: Mrs. Leah Martha Mercanti
Last Name of Student: Ferro First Name: Demetrio Place and date of birth: Sapri, SA – 02 December 1990 Sex : M E-mail address: ferrodemetrio@gmail.com Notes: with scholarship

Degree

Name of the qualification	PhD in Cognitive and Brain Sciences
Specialization	Cognitive Neuroscience
PhD Thesis title	Effects of attention on visual processing between cortical layers and cortical areas V1 and V4
PhD Thesis Supervisor	Stefano Panzeri
Thesis Defense date	December 13, 2019
Language of instruction and research activity	English

Information about the degree level

Level of the qualification	Doctorate of Philosophy
Required credits	180 credits
Entrance requirements	Master's degree (<i>Laurea Magistrale</i>) in Telecommunications Engineering
Mode of study	Full time



Programme's start date	November, 2016
Duration	3 years

Acknowledged credits

EDUCATION	Doctoral courses, seminars, conferences*	60
RESEARCH	Literature review, laboratory and field research, writing of thesis	120
EXAM	Thesis discussion	pass
Total amount of credits		180

Courses in detail

COURSE NAME	STATUS	CREDITS	YEAR
Make the most of your PhD	Pass	1	1
Ethics of research in Neuroscience	Pass	2	1
RC1 -Data visualization	Pass	1	1
RC2 – Figures and posters	Pass	1.5	1
RC3 – Presentations	Pass	2	1
RC5 How to review a journal article	Pass	1	1
Methods 3: MEG (Magnetoencephalography)	Pass	1.5	1
Methods 4: TBS (Transcranial Brain Stimulation)	Pass	1.5	1
Advanced Statistics	Pass	2	1
Funding your research	Pass	0.5	1
Colloquia attendance	Pass	7.5	1,2,3
Brown Bag attendance	Pass	6	1,2,3
Doctoral Student Day attendance	Pass	2	1,2



Participation	Pass	1.5	1,2,3
Total certifiable course credits (8 hr. frontal time per credit)		31	

Information about the Doctorate Programme in Cognitive and Brain Sciences

The 3-year doctorate in Cognitive and Brain Sciences has an academic level and a duration comparable to PhD courses given at other universities in Europe and worldwide.

Enrolment: on a selective basis; scholarships are assigned only to the best shortlisted candidates.

Grading: on the basis of the Italian national regulations, there is no grading system for doctoral students.

Credits: to complete the whole programme, the student must obtain 180 credits.

Degree award: the doctoral degree is awarded to doctoral students who have completed the requested educational and research activity (obtaining 180 credits) and have written an original and innovative thesis, after its defence before an international examination committee.

Further information on the School website: <http://www.unitn.it/drcimec>.

I hereby certify that **Demetrio Ferro** attended all 3 years of the Doctorate in Cognitive and Brain Sciences of the Center for Mind/Brain Sciences (CiMeC), University of Trento, Italy. On 9 December, 2019 he successfully defended his PhD thesis. The **PhD's Oral Defense Commissioners** were (President) Prof. Demian Battaglia, Prof. Alessandro Treves and Prof. Paul Hamilton.

Date	Signature of registrar	Stamp of institution
4 Feb. 2020		 UNIVERSITÀ DEGLI STUDI DI TRENTO



DOCTORATE IN COGNITIVE AND BRAIN SCIENCES

This is to certify that Dr. Demetrio Ferro participated in the following conference activities during his PhD from November 1, 2016 until December 13, 2019:

1	Authors (in order of appearance) – Underline your name	Thiele A., <u>Ferro D.</u> , Boyd M., Panzeri S.
	Title of the work presented	Layer dependent attentional modulation of broad and narrow spiking cells in primate V1
	Type of conference (international, national, etc.)	Internacional
	Name of the conference	<i>Annual Meeting of the Society for Neuroscience (SfN)</i>
	Dates of the conference	2017
	Venue/location of the conference	Washington, DC, USA
	Type of participation (oral presentation, poster, etc.)	Poster
	Remarks (if applicable)	
2	Authors (in order of appearance) – Underline your name	<u>Ferro D.</u> , van Kempen J., Boyd M., Panzeri S., Thiele A.
	Title of the work presented	Effects of attention on visual processing between cortical layers and cortical areas V1 and V4
	Type of conference (international, national, etc.)	Internacional
	Name of the conference	<i>Ten Years of Mind/Brain Sciences Conference at the University of Trento</i>
	Dates of the conference	2017
	Venue/location of the conference	Rovereto, Italy
	Type of participation (oral presentation, poster, etc.)	Poster
	Remarks (if applicable)	
3	Authors (in order of appearance) – Underline your name	<u>Ferro D.</u> , van Kempen J., Boyd M., Panzeri S., Thiele A.



UNIVERSITÀ
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CiMeC
Center for Mind/Brain Sciences
Doctoral Programme in Cognitive and Brain Sciences

Title of the work presented	Effects of attention on visual processing between cortical layers and cortical areas V1 and V4
Type of conference (international, national, etc.)	Local
Name of the conference	<i>PhD Doctoral Student Day, University of Trento</i>
Dates of the conference	2018
Venue/location of the conference	Rovereto, Italy
Type of participation (oral presentation, poster, etc.)	Poster
Remarks (if applicable)	

4	Authors (in order of appearance) – Underline your name	<u>Ferro D.</u> , van Kempen J., Boyd M., Panzeri S., Thiele A.
	Title of the work presented	Effects of attention on Granger causal interactions between cortical layers and cortical areas V1 and V4
	Type of conference (international, national, etc.)	Internacional
	Name of the conference	<i>Annual Meeting of the Society for Neuroscience (SfN)</i>
	Dates of the conference	2019
	Venue/location of the conference	Chicago, IL, 2019
	Type of participation (oral presentation, poster, etc.)	Poster
	Remarks (if applicable)	

Date	Signature of registrar	Stamp of institution
June 4, 2025	LEAH M. MERCANTI 	 UNIVERSITÀ DI TRENTO CiMeC Center for Mind/Brain Sciences

CERTIFICAT DE DECLARACIÓ D'EQUIVALÈNCIA AL NIVELL ACADÈMIC DE DOCTOR

CERTIFICADO DE DECLARACIÓN DE EQUIVALENCIA AL NIVEL ACADÉMICO DE DOCTOR

De conformitat amb el que es disposa en el Reial decret 889/2022, de 18 d'octubre, pel qual s'estableixen les condicions i els procediments d'homologació, de declaració d'equivalència i de convalidació d'ensenyaments universitaris de sistemes educatius estrangers i pel qual es regula el procediment per a establir la correspondència al nivell del Marc Espanyol de Qualificacions per a l'Educació Superior dels títols universitaris oficials pertanyents a ordenacions acadèmiques anteriors (BOE núm. 251, de 19 d'octubre de 2022), el Rector Magnífic de la Universitat Politècnica de Catalunya ha resolt:

Que el títol de **DOTTORE DI RICERCA IN COGNITIVE AND BRAIN SCIENCES** obtingut per **DEMETRIO FERRO**, nascut el 2 de desembre de 1990, de nacionalitat italiana, a la **UNIVERSITÀ DEGLI STUDI DI TRENTO (ITÀLIA)**, té un nivell acadèmic equivalent al del títol universitari oficial espanyol de **Doctor per la Universitat Politècnica de Catalunya**.

En el seu efecte, aquesta Universitat expedeix el present certificat, que acredita els efectes d'aquesta equivalència des de la seva data d'expedició.

Barcelona, 24 de març de 2026

El Vicerector de Recerca
Pedro Díez Mejía
Per delegació de competència del rector
Resolució 122/2020, de 28 de gener, de delegació de competències (DOGC Núm. 8082 d'11.3.2020)

2026/Q04205
(Núm. d'inscripció a la secció especial del
Registre Nacional de Títols)

De conformidad con lo dispuesto en el Real Decreto 889/2022, de 18 de octubre, por el que se establecen las condiciones y los procedimientos de homologación, de declaración de equivalencia y de convalidación de enseñanzas universitarias de sistemas educativos extranjeros y por el que se regula el procedimiento para establecer la correspondencia al nivel del Marco Español de Cualificaciones para la Educación Superior de los títulos universitarios oficiales pertenecientes a ordenaciones académicas anteriores (BOE núm. 251, de 19 de octubre de 2022), el Rector Magnífico de la Universitat Politècnica de Catalunya ha resuelto:

*Que el título de **DOTTORE DI RICERCA IN COGNITIVE AND BRAIN SCIENCES** obtenido por **DEMETRIO FERRO**, nacido el 2 de diciembre de 1990, de nacionalidad italiana, en la **UNIVERSITÀ DEGLI STUDI DI TRENTO (ITALIA)**, tiene un nivel académico equivalente al del título universitario oficial español de **Doctor por la Universitat Politècnica de Catalunya**.*

En su virtud, esta Universidad expide el presente certificado, que acredita los efectos de dicha equivalencia desde su fecha de expedición.

Barcelona, 24 de marzo de 2026

El Vicerector de Investigación
Pedro Díez Mejía
Por delegación de competencia del rector
Resolución 122/2020, de 28 de enero, de delegación de competencias (DOGC Núm. 8082 de 11.3.2020)

2026/Q04205
(Núm. de inscripción en la sección especial del
Registro Nacional de Títulos)

Original signat per:

PEDRO DIEZ MEJIA
Universitat Politècnica de Catalunya
Signat en: 24-03-2026 09:35:38
GMT+1



Demetrio Ferro

Centre de Recerca Matemàtica, Edifici C, 080193, Bellaterra, Barcelona, Spain

Email: dferro@crm.cat, ferrodemetrio@gmail.com, Website: d-ferro.github.io

Researcher ID: [JCE-8721-2023](#), Scopus ID: [57221351783](#), ORCID ID: [0000-0003-4969-1415](#).

SUMMARY I am a computational neuroscientist working at the interface of neural data analysis and theory. My research focuses on how attention, gaze, and internal goals shape decision-making and action. I have developed information-theoretic and spectrally resolved methods to quantify directed, laminar-specific interactions in visual cortical circuits, as well as computational methods showing how gaze and goal states dynamically modulate value encoding in prefrontal brain regions. More recent works focus on theoretical modelling of neural population firing rates for the integration of sensory information and gaze coordination in decision-making. My work combines behavioral and electrophysiological data analyses, advanced statistical signal processing, and theoretical modeling to link circuit mechanisms to neural coding and cognitive function across scales.

EDUCATION **PhD in Cognitive, Computational Neuroscience** | [Diploma supplement](#)
University of Trento, Center for Mind/Brain Sciences, Rovereto, Italy
Istituto Italiano di Tecnologia, Center for Neuroscience and Cognitive Systems, Rovereto, Italy
Defense date: 13 Dec 2019; Supervisor: Prof. Stefano Panzeri.

My PhD project was conducted in collaboration with the experimental lab of Prof. Alexander Thiele (University of Newcastle, UK). I studied electrophysiological recordings from visual cortical areas V1 and V4 to characterize directed causal interactions across laminar layers within and between the two areas. I developed information detection methods combining information theory, spectral analysis, and statistical signal processing, introducing autoregressive metrics of spectral and conditional causality based on mutual information. This work enabled a laminar- and frequency-resolved characterization of information flows, showing spectral segregation in feedforward and feedback interactions and strong modulation of neural signals and interactions by visuospatial attention ([Ferro et al., PNAS, 2021](#)). All computational methods developed during the project were made publicly available, including tools for [laminar alignment via current source-density](#) and [neural response latency](#). This work was presented in multiple international conferences (*SfN* 2017, 2019) and in talks invited by peer scientists conducting research in the field (at Institut de Neurosciences de la Timone, Marseille, France in 2020; and at University of Amsterdam, Netherlands, in 2024).

M.Sc. degree in Telecommunications Engineering (“cum laude”) | [Diploma supplement](#)
Polytechnic of Turin, Dept. of Electronics and Telecommunications (DET), Turin, Italy
Dissertation: 11 Dec 2015; Thesis supervisor: Prof. G. Montorsi (DET),
External supervisors: Prof. C. Berrou and Prof. V. Gripon (Télécom Bretagne, Brest, France).

I acquired solid foundations in stochastic signal processing, system programming, wireless and optical communications, and carried several projects about [Information Theory](#), [Coding Theory](#), [Communication systems](#), [Queueing theory](#), [Cryptography](#), [Satellite navigation](#), [Remote sensing](#). My master degree thesis was conducted through a 6-month internship at Télécom Bretagne (Brest, France), in the lab of [Prof. Claude Berrou](#), pioneer in the field of Information Theory, supervised by Prof. V. Gripon. The thesis consisted in developing neuro-inspired neural network methods for efficient nearest-neighbor search in high-dimensional data, achieving ~95% reduction for classification complexity and ~77% reduction for image search ([IJCNN, Vancouver 2016](#)).

B.Sc. degree in Computer Engineering | [Diploma supplement](#)
University of Salerno, Department of Information Engineering, Electrical Engineering, and Applied Mathematics, Italy; Dissertation: 23 Sept 2013; Supervisors: Prof. S. Marano and Prof. V. Matta

I built a solid background in mathematics, physics, statistics, electronics, digital circuit modeling, programming (C/C++, Assembly, Java, MATLAB, MATLAB GUI, Python, SQL, JavaScript, HTML), computer architecture, logical circuit design, system programming, operating systems (Microsoft Windows, Linux), telecommunications and network protocols (SSH, SFTP, RDP, etc.).

My bachelor thesis project was conducted at Co.Ri.Tel (Ericsson) and it focused on analyzing information flows hidden under cover traffic. I developed algorithms to detect covert traffic flows and derived theoretical limits for the detectable channel capacity by modelling traffic as Poisson point processes. Both results were extended to bidirectional flows and multi-hop routing.

POSITIONS
COVERED

01/12/2025 – current | Postdoctoral Researcher

Centre de Recerca Matemàtica, Campus Bellaterra, Universitat Autònoma de Barcelona

Postdoctoral researcher in the lab of Prof. Klaus Wimmer. In collaboration with Prof. Alexander Huk (UCLA), my research focuses on the neural mechanisms of perceptual decision-making, combining experimental data from motion direction discrimination tasks with computational modeling. I developed a two-population recurrent network model linking sensory and decision circuits, reproducing adaptive temporal weighting of sensory evidence (primacy, uniform, and recency strategies) observed in behavior and neural activity. This work strengthens my expertise in theoretical modelling, differential network dynamics, and simulation-based inference, providing mechanistic insight into how contextual signals shape decision-making in neural circuits.

11/10/2020 – 30/11/2025 | Postdoctoral Researcher

Center for Brain and Cognition, Department of Engineering, Universitat Pompeu Fabra (UPF) in Barcelona, Spain

In the lab of Prof. Rubén Moreno-Bote, I investigated value-based decision-making and the influence of internal goals on neural evaluation processes. In collaboration with Prof. Benjamin Hayden (Baylor College of Medicine, USA), I developed a framework to study how directing gaze toward the spatial location of reward-associated cues selectively modulates their neural value representations in orbitofrontal cortex, both during their presentation and at later blank screen delays before choice, and that modulation strength is choice-predictive (*Nature Communications*, 2024). Furthermore, I investigated the role of goal-directed influences on decision-making. Using a token-based task, I found that subjects made faster and more accurate decisions as the accumulation of tokens brought them closer to a jackpot reward. By modeling subjective value through a utility function inspired by Prospect Theory, I characterized a reference-shifting strategy whereby value is evaluated as potential gain or loss depending on reward proximity. This study showed that dorsal anterior cingulate cortex enhances the neural encoding of potential gains relative to losses, through the recruitment of larger encoding populations, and that neural value encoding weights are tightly linked to choice behavior (*Nature Communications*, 2026). Through collaboration with Prof. Tianming Yang (NYU-ECNU, Shanghai), I developed a closed-loop computational model linking visual sampling and decision formation (*CCN* 2025), which I am currently integrating with experimental data. In parallel, I collaborate with Prof. Sonja Grün (Jülich, Germany) and Dr. Thomas Brochier (Marseille, France) on context-dependent neural encoding, investigating how individual motor actions recruit specialized neural representations across overlapping action sequences. Overall, my postdoctoral work broadened my expertise across attention, intention, decision, and action, combining neural modeling with gating mechanisms to explain how functional states shape sensory and motor computations.

01/11/2019 – 31/08/2020 | Research assistant at the Center for Neuroscience and Cognitive Systems, Istituto Italiano di Tecnologia (Rovereto, Italy)

During this period, I consolidated and extended my PhD research, leveraging the richness of the experimental dataset to pursue additional analyses and broaden the scope of investigation. This work contributed to strengthening my profile and supported the completion and dissemination of

my PhD work. This time overlapped with COVID-19 pandemic, which delayed the start of my postdoc.

My tasks focused on getting all the materials and methods produced during my PhD ready for open-access publication in my own repositories (<https://gin.g-node.org/56fe>) and to work on manuscript writing and reviews, later accepted for publication (Ferro et al., *PNAS* 2021).

01/11/2016 – 31/10/2019 | PhD Candidate at the Center for Mind/Brain Sciences, University of Trento (Rovereto, Italy)

Branching from engineering studies, I pursued a PhD project in the computational lab of Prof. Stefano Panzeri in collaboration with the experimental lab of Prof. Alexander Thiele in Newcastle, UK. We faced multiple experimental and theoretical challenges but exploited our knowledge to combine the efforts in developing robust and efficient tools for achieving the quality assessment and anatomical alignment of high-resolution laminar neural data of our interest. The initial part of this project expanded broadly due to the complexity of the methods but enriched my background and contributed to my current knowledge of primate neurophysiology. The work was published (Ferro, *IRIS* 2019; Ferro et al., *PNAS* 2021) along with multiple repositories for laminar alignment, latency estimation and directed information flows detection (<https://gin.g-node.org/56fe>).

01/04/2016 – 30/10/2016 | Internship at the Center for Neuroscience and Cognitive Systems, Istituto Italiano di Tecnologia (Rovereto, Italy)

I embarked on my neuroscience journey, thanks to the support of Prof. Stefano Panzeri and his research group. This time was particularly fruitful for me to approach research life and to acquire study certificates for highly relevant methods (Machine Learning by Prof. Andrew Ng) and the basis of neuroanatomy and neural functions (Medical Neuroscience by Prof. Leonard E. White).

01/04/2015 – 30/09/2015 | Master thesis at Institut Mines-Télécom (Brest, France)

I developed a novel solution for nearest neighbor search, a common task in computer vision and pattern recognition. The method combined vector quantization with neural networks to outperform state-of-the-art in efficiency and scalability, achieving about 95% complexity reduction in image classification, and 77% for image search. Presented and published in *IJCNN, Vancouver 2016*.

TEACHING
MENTORING

Teaching

I taught Scientific Communication (3 ECTS) in the bachelor's degree of Biomedical Engineering. Modules: I. Scientific writing and source referencing; II. Public speech evaluation and presentation delivery; III. Reviewing essays of peer students and proposing alternative versions fitting instructed guidelines; IV. Reproducibility and transparency of publication, the licenses for software and data, repositories and digital object identifiers. Students were evaluated by a weighted sum of four assignments, one for each module. For module II, each student was asked to prepare an individual presentation and evaluated individually with point-by-point feedback. For this activity, I obtained a Teaching certificate by the university and positive evaluation feedback by the students.

I have been **chair of the committee** for **bachelor's** theses in 2024–2026, and for **master's** theses in the Neuroscience panel in 2022 at UPF. In 2026, I have been **chair** and **referee** of the **PhD thesis** committee for 2 PhD defenses at SISSA (Trieste, Italy) and **referee** of 1 PhD thesis at UPF.

Mentoring

I supervised 2 bachelors theses (6 months each) and 2 internship students (3 months each), as formally certified by the Engineering Department at Universitat Pompeu Fabra.

The first thesis student (2020–2021) analyzed spiking activity to study the neural encoding of task-related cognitive variables in a decision-making task, contributing to Ferro et al, *CCN* (2023).

The second thesis student (2024) worked on Allen Institute's "[Visual Behavior Neuropixels](#)" data, implementing spectral and laminar analyses to classify signals by cortical layers and task conditions, to be integrated in my current research framework for publication planned in future work. The student later joined a top-tier computer science master program in Japan.

Internship students worked with publicly available data from the Allen Brain Institute. The first student characterized covariability and mutual information across neural populations, while the second student analyzed response latencies across laminar depths and task conditions. The first student continued into a prestigious computer science program, and the second, inspired by the internship, pursued a thesis on brain networks and enrolled in a competitive Computational Neuroscience master at the ETH in Zürich, which I gladly supported via recommendation letter.

More recently, I have supervised 1 PhD student, collaborated at project development and substantially supporting methodological rigor, project structure and thesis writing.

RESEARCH PROJECTS

Neural network mechanisms underlying perceptual decision-making (2025 – ongoing)

Collaboration with the lab of [Prof. Alex Huk](#) at UCLA, in the USA. Neural network models of sensory and decision-formation neural populations for adaptive temporal-weighting of sensory information (recency, primacy) in a perceptual motion discrimination task (under development).

Gaze2Decision (2024 – ongoing)

Collaboration with the experimental lab of [Dr. Tianming Yang](#) at the NYU-ECNU in Shanghai, China. Closed-loop circuit modelling of two-paired mutually inhibiting networks of neural units for the interaction between gaze direction and decision formation ([Ferro et al., CCN 2025](#)).

Vision4Action (2024 – ongoing)

Joint venture with the lab of [Prof. Sonja Grün](#) (Institute for Advanced Simulations, Jülich, Germany) and the experimental lab of [Dr. Thomas Brochier](#) (Institut de Neurosciences de la Timone, Marseille, France). Analysis of neural data collected during a hand landing task for the coordination of visual and motor functions (under development).

Hayden Lab (2021 – ongoing)

Collaboration with the experimental research lab of [Prof. Benjamin Hayden](#) at the Baylor College of Medicine, Houston, USA. Analysis of the neural encoding of reward-based values and the role of gaze position in decision-making tasks ([CCN 2023](#); [Nature Communications 2024](#); [CCN 2025](#); [Nature Communications 2026](#)).

Human Brain Project (HBP), Co-funded by the EU (2022-2023)

Consortium of Research on Decision-Making (CoReDeM), bringing together theoretical (PIs: [R. Moreno-Bote](#) and [I. Cos](#) at Univ. Pompeu Fabra, Barcelona; [A. Destexhe](#) from Centre National de la Recherche Scientifique Paris Saclay) and experimental research groups (PIs: [S. Ferraina](#), Sapienza University, Rome, and [T. Theys](#), KU Leuven, Belgium). [[Report: SGA3 M42](#); [EBRAINS](#)].

The effects of attention on laminar interactions (2017 – 2021)

Collaboration with the experimental group of [Prof. A. Thiele](#) at the University of Newcastle, UK. Analysis of laminar-specific effects of attention on inter- and intra-areal directed information flows within brain areas V1 and V4 ([Ferro, IRIS 2019](#); [Ferro et al., PNAS 2021](#)).

PUBLICATIONS **Pre-prints**

"Dynamic interactions between sensory and decision-making circuits shape temporal evidence weighting"

D. Ferro*[✉], C. Vivar*, A. J. Levi, A. C. Huk and K. Wimmer

bioRxiv, 2026; [PDF](#); [✉]**Corresponding author**, *Equal contribution.

Adaptive perceptual decisions depend highly on flexibly weighting sensory evidence as environmental cues change over time. We developed a psychophysiological grounded computational model that reproduces neural and behavioral signatures of temporal evidence weighting, identifying mechanistic circuit interactions through which sensory inputs and top-down signals dynamically shape decisions.

Peer-reviewed journal articles

Nature Communications 2026

"Accumulation of virtual tokens towards a jackpot reward enhances performance and value encoding in dorsal anterior cingulate cortex"

D. Ferro[✉], H. Azab, B. Y. Hayden and R. Moreno Bote

Nature Communications, 17:2537, 2026; [10.1038/s41467-026-70423-1](https://doi.org/10.1038/s41467-026-70423-1); [✉]Corresponding author.

This publication was an **invited submission by Senior Editor**. The work has now been **featured in Nature Portfolio Collection**: *"Neural mechanisms underlying sensory-guided motor control"*.

In proximity of large rewards, subjects perform choices faster and more accurately, reflecting a fundamental shift in their evaluation strategy that is not readily explained by speed versus accuracy trade-off. The evaluation of alternative options is referenced to a cumulative token count leading to a large jackpot reward. Neural activity in dorsal anterior cingulate cortex tracked token reference shifts, revealing a neural basis for reference-dependent evaluation in decision-making.

Nature Communications 2024

"Gaze-centered gating, reactivation, and reevaluation of economic value in orbitofrontal cortex"

D. Ferro[✉], T. Cash-Padgett, M. Zhe-Wang, B. Y. Hayden, R. Moreno Bote,

Nature Communications, 15:6163, 2024; [10.1038/s41467-024-50214-2](https://doi.org/10.1038/s41467-024-50214-2);

[✉]Corresponding author; Citations: 14 (Scholar, Sept. 2026).

Gaze behavior strongly influences decision-making during economic choices. Looking at the screen locations where options are or were presented activates or reactivates their value encoding in orbitofrontal cortex. This reactivation correlates with upcoming choices, indicating that gaze plays an active role in reevaluating internal representations.

PNAS 2021

"Directed information exchange between cortical layers in macaque V1 and V4 and its modulation by selective attention"

D. Ferro, J. van Kempen, M. Boyd, S. Panzeri[✉], A. Thiele[✉]

Proceedings of the National Academy of Sciences (PNAS), 118 (12), e2022097118, 2021

[10.1073/pnas.2022097118](https://doi.org/10.1073/pnas.2022097118); [✉]Corresponding authors; Citations: 70 (Scholar, Sept. 2026).

This work maps how visual and attentional information propagate within and between cortical areas V1 and V4 at laminar resolution. Results show interlaminar and inter-areal circuit motifs, attention-related shift in the high-frequency band spectral peak and increased high-frequency power. Attention reduced local communication within V1 but enhanced both feedforward and feedback signaling across areas, especially in the high-frequency bands.

Peer-reviewed Conference Proceedings

CCN 2025

"A closed-loop model for the coordination of gaze control and decision-making"

D. Ferro[✉], T. Yang and R. Moreno-Bote,

Cognitive, Computational Neuroscience (CCN), 2025; [PDF](#); [✉]Corresponding author.

This work introduces a closed-loop model linking gaze behavior with value-based decision-making. Visual input drives competing decision populations, which guide gaze shifts that feed back into the

decision process. Simulations capture key behavioral dynamics, offering a framework to study how gaze shapes neural and decision-related activity.

CCN 2025

"Decision-making reference point biases in the dorsal anterior cingulate cortex"

D. Ferro✉, H. Azab, B. Y. Hayden, R. Moreno-Bote,
Cognitive, Computational Neuroscience (CCN) 2025; [PDF](#); ✉Corresponding author.

This study examined how reference-point bias affects probabilistic decision-making in a behavioral task where virtual tokens indicate jackpot achievement odds. As token counts increased, participants made quicker, more accurate decisions, reflecting sensitivity to current reward status. Performance also improved in easier trials with more salient, high-value options, suggesting reference-dependent biases influence behavior beyond the standard speed-accuracy tradeoff.

CCN 2023

"The role of gaze for value encoding and recollection in orbitofrontal cortex"

D. Ferro✉, A. Rifé-Mata, T. Cash-Padgett, M. Zhe-Wang, B. Hayden, R. Moreno-Bote
Cognitive, Computational Neuroscience (CCN), 2023; [PDF](#); ✉Corresponding author.

Gaze is drawn more to valuable options and predicts choice outcomes. Orbitofrontal cortex activity enhances the encoding for fixated locations and suppression of alternatives. These findings suggest that gaze supports internal deliberation by modulating neural value signals.

IJCNN 2016

"Nearest neighbour search using binary neural networks"

D. Ferro✉, V. Gripon, X. Jiang,
International Joint Conference on Neural Networks (IJCNN), IEEE, 5106-5112, 2016
[10.1109/IJCNN.2016.7727873](#); Citations: 12 (Scholar, Sept. 2026); ✉Corresponding author.

Nearest neighbor search is a common task in computer vision and pattern recognition whose complexity grows with data dimensionality. The method proposed here combines Product Quantization with neural networks to outperform state-of-the-art methods both in efficiency and scalability, achieving about 95% reduction in classification on [MNIST dataset](#), and 77% reduction in nearest neighbor search on 1M image descriptors dataset ([TEXMEX 1M](#), INRIA, Rennes, France).

PEER REVIEWS

I have contributed multiple times to reviewing the work of peer authors. As of 2026:

- 1 peer reviews for Nature Neuroscience (verifiable at [ORCID](#)),
- 1 peer reviews for Nature Communications (verifiable at [ORCID](#)),
- 7 peer reviews for PLOS Computational Biology (verifiable at [Web of Science](#)),
- 1 peer review for Qeios (verifiable at [Web of Science](#)),
- 20+ abstracts reviewed for [Cosyne 2026–2027](#), 6+ abstracts reviewed for [CCN 2023](#), [CCN 2025](#).

EVENTS ORGANIZATION

In the last years, I have taken part in and organized multiple social events within the network of scientists and neuroscientists in Barcelona. I joined and actively participated in the [Barcelona Computational, Cognitive and Systems Neuroscience community](#). I participated in annual conferences, seminars, presented my work multiple times, and organized two research retreats for the whole network (in 2022 and in 2024) including members at all career stages (~60 attendees). The retreats consisted of 3 days of reunions of the network with fully immersive scientific and community-related discussions, and great opportunities for mutual exchange and networking.

During my PhD, (March 2017 – April 2018) I organized a year-long series of weekly seminars inviting local and visiting scientists interested in sharing and promoting their research, highly relevant for the local community of the Center for Mind/Brain Sciences, Rovereto, Italy.

In addition, I have established connection with the Societat Catalana de Biologia (SCB) and the Centre de Recerca Matemàtica (CRM), where I met peer members with whom I enjoyed great conversations. Attracted by the "Mes Cajal" events organized by the SCB, I also started to interact

with colleagues involved in the research fields of biotechnology, oncology and cell cultures. Through their invitation, I joined the "PostDoc Day in Biomedical research", co-organized by SCB, and extended invitation to neuroscience colleagues, proudly connecting people across fields.

Similarly, during the last years I interacted on many occasions with biomedical and computational sciences colleagues ([IDIBAPS](#) at Hospital Clinic, [IRB](#) and [VHIQ](#), [IMIM](#) and [PRBB](#) at Hospital de Mar, [UB](#), [UAB](#) and at the [CRM](#)). I joined the CRM as a postdoc in December 2025, through their series of "Neurochats", bringing together pre- and post-doctoral researchers in periodic meetings where I also had the chance to present in May 2025. In addition, I was invited to present my work to the hub of "[Computational and Mathematical Neuroscience](#)", invited by Dr. A. Hyafil and lectured at [PRUAB](#) for the [TOUCH](#) project ([MSCA-COFUND-101126533](#)), invited by Dr. J. Malapeira.

PUBLIC OUTREACH

Since the beginning of my career, I have been committed to scientific communication, with the clear idea that scientific discoveries lose their impact if they have poor outreach or if they are only accessible by a minority of the population. I have spent many years of my training in studying open access policies, developing open-source software, and learning how to properly publish data and code in a transparent, reproducible way, as by the standards of findability, accessibility, interoperability, and reusability (FAIR). All my research is published on open access journals, and all the data and code are publicly available on my [repository page](#).

I have enthusiastically participated in public outreach events ([European Researcher's Night](#), [Festa de la Ciència](#) and [Pint of Science](#)) in Barcelona, and previously, in France and in Italy.

I have published informative articles about my research in social media (Linkedin, Bluesky) and on important scientific outlets like [EurekAlert \(AAAS\)](#), [Medical Xpress \(Science X\)](#), [Bioengineer.org](#) and [News-Medical Life Sciences](#). I received multiple interviews by scientific communication press featured in UPF News [2021](#), [2024](#), [2026](#) and in CRM News 2026.

INTERNATIONAL VISIBILITY

Contributed Presentations in the form of posters (p) or talks (t). Among the most relevant:

- (t, p) Italian Network for Computational and Theoretical Neuroscience, Sept. 2026 (Trieste, Italy).
- (p) Federation of European Neuroscience Societies ([FENS](#)), July 2026 (Barcelona, Spain).
- (p) Barcelona Cognitive, Computational Systems Neuroscience, May 2026 (Barcelona, Spain).
- (p) Systems in Action, March 2026 (Alicante, Spain).
- (p) Cognitive, Computational Neuroscience, August 2025 (Amsterdam, Netherlands).
- (t) Citizen Art-Science Engagement Strategies (CASES), July 2025 (Barcelona, Spain).
- (t, p) International Conference on Thinking, June 2024 (Milan, Italy).
- (p) Cognitive, Computational Neuroscience, August 2023 (Oxford, UK).
- (p) Human Brain Project Summit, March 2023 (Marseille, France).
- (p) Barcelona Cognitive, Computational Systems Neuroscience, May 2022 (Barcelona, Spain).
- (p) Cognitive Systems Neuroscience (Cosyne), March 2022 (Lisbon, Portugal).
- (t) CoReDeM (Human Brain Project), Nov. 2021 (Barcelona, Spain)
- (t) Spanish Network for the Interaction between Computational and Cognitive Neuroscience (SINC²), Nov. 2021 (Lleida, Spain).
- (p) Spanish Society for Neuroscience (SENC), Nov. 2021 (Lleida, Spain).
- (p) Barcelona Cognitive, Computational Systems Neuroscience, July 2021 (Spain).
- (p) Annual Meeting of the Society for Neuroscience (SfN), 2019 (Chicago, USA).
- (p) PhD Doctoral student Day at the University of Trento, 2019 (Rovereto, Italy).
- (p) Ten Years of Mind/Brain Sciences, University of Trento, 2017 (Rovereto, Italy).
- (p) Annual Meeting of the Society for Neuroscience (SfN), 2017 (Washington, USA).

You can find an [exhaustive, updated list](#) of all conferences on my website.

Invited talks as a speaker at recurrent occasions, approximately annually, presenting my research findings and disseminating methods I developed and made publicly available to peers in the field.

- 2026 Citizen Art-Science Engagement Strategies (CASES) 2026 (Turin, Italy)

- 2026 Parc de Recerca de la Universitat Autònoma de Barcelona (Barcelona, Spain)
- 2025 Centre de Recerca Matemàtica (Barcelona, Spain)
- 2024 Universiteit van Amsterdam (Amsterdam, Netherlands)
- 2023 Center for Mind/Brain Sciences (Rovereto, Italy)
- 2023 Human Brain Project, Work Package 2 Meeting (Barcelona, Spain)
- 2022 Multi-Scale Brain Function India-Italy Network of Excellence (Varenna, Italy)
- 2020 Institut de Neurosciences de la Timone (Marseille, France)

AWARDS

Funding

I conceived, wrote, submitted, and was awarded funding for a research proposal fully conceived and written by me on “gaze-centered decision-making,” supporting my 2023–2025 postdoctoral research activities, awarded in 2022 by Bial Foundation ([PT/FB/BL-2022-106](#)). I also led the curation of progress reports and the [final project report](#).

OTHER ACTIVITIES

Citizen Science

I actively participate in the Citizen Art-Science Engagement Strategies (CASES) network, operating within the EUTOPIA Impact and Dissemination programme at the Universitat Pompeu Fabra in Barcelona, a collaborative alliance of like-minded European universities. The initiatives include collective environmental data sampling, informational meetings on scientific reports. I have contributed by disseminating my research on attention, decision-making and implications for executive functions and impairments to network members in [2025](#) and to school students in [2026](#).